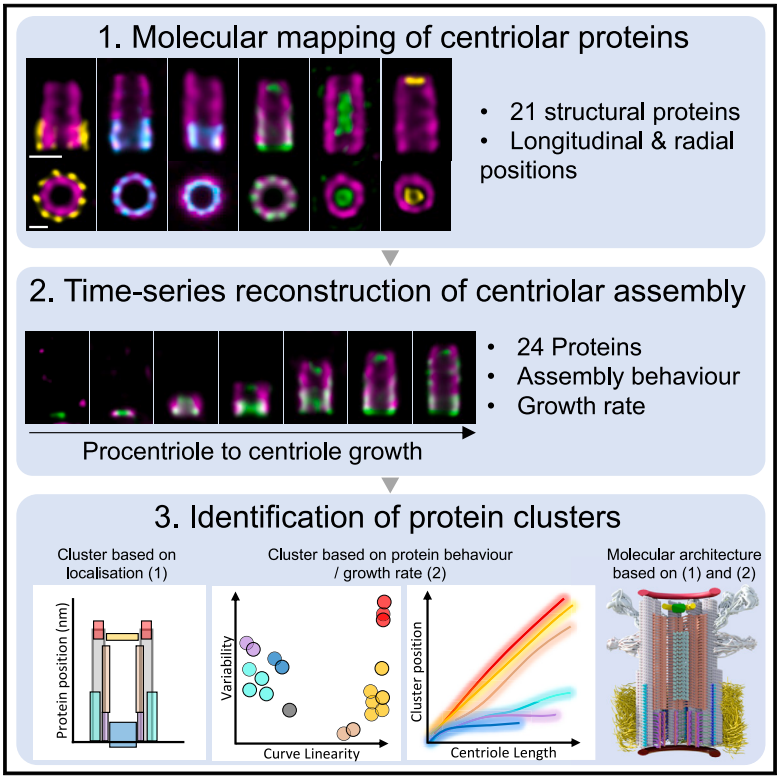
Cell

Time-series reconstruction of the molecular architecture of human centriole assembly

Graphical abstract



Highlights

- Expansion microscopy reveals the molecular architecture of centriole assembly
- Dissection of the behavior of 24 centriolar proteins
- Cluster analysis reveals six sub-structural elements
- Modeling of the architectural growth of the human centriole

Authors

Marine H. Laporte, Davide Gambarotto, Éloïse Bertiaux, ..., Susanne Borgers, Virginie Hamel, Paul Guichard

Correspondence

virginie.hamel@unige.ch (V.H.), paul.guichard@unige.ch (P.G.)

In brief

Expansion microscopy and time-series reconstructions enable elaboration of dynamic centriole assembly.

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Time-series reconstruction of the molecular architecture of human centriole assembly

Marine H. Laporte,[1,3] Davide Gambarotto,[1,4] Éloïse Bertiaux,[1] Lorène Bournonville,[1] Vincent Louvel,[1] José M. Nunes,[2] Susanne Borgers,[1] Virginie Hamel,[1,*] and Paul Guichard[1,5,*]

[1]University of Geneva, Department of Molecular and Cellular Biology, Faculty of Sciences, Geneva, Switzerland
[2]University of Geneva, Department of Genetic and evolution, Faculty of Sciences, Geneva, Switzerland
[3]Present address: University of Lyon, Université Claude Bernard Lyon 1, MeLiS, UCBL, CNRS UMR 5284, INSERM U1314, Institut NeuroMyoGène, Lyon 69008, France
[4]Present address: Viventis Microscopy Sàrl, Lausanne, Switzerland
[5]Lead contact
*Correspondence: virginie.hamel@unige.ch (V.H.), paul.guichard@unige.ch (P.G.)
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SUMMARY

Centriole biogenesis, as in most organelle assemblies, involves the sequential recruitment of sub-structural elements that will support its function. To uncover this process, we correlated the spatial location of 24 centriolar proteins with structural features using expansion microscopy. A time-series reconstruction of protein distributions throughout human procentriole assembly unveiled the molecular architecture of the centriole biogenesis steps. We found that the process initiates with the formation of a naked cartwheel devoid of microtubules. Next, the bloom phase progresses with microtubule blade assembly, concomitantly with radial separation and rapid cartwheel growth. In the subsequent elongation phase, the tubulin backbone grows linearly with the recruitment of the A-C linker, followed by proteins of the inner scaffold (IS). By following six structural modules, we modeled 4D assembly of the human centriole. Collectively, this work provides a framework to investigate the spatial and temporal assembly of large macromolecules.

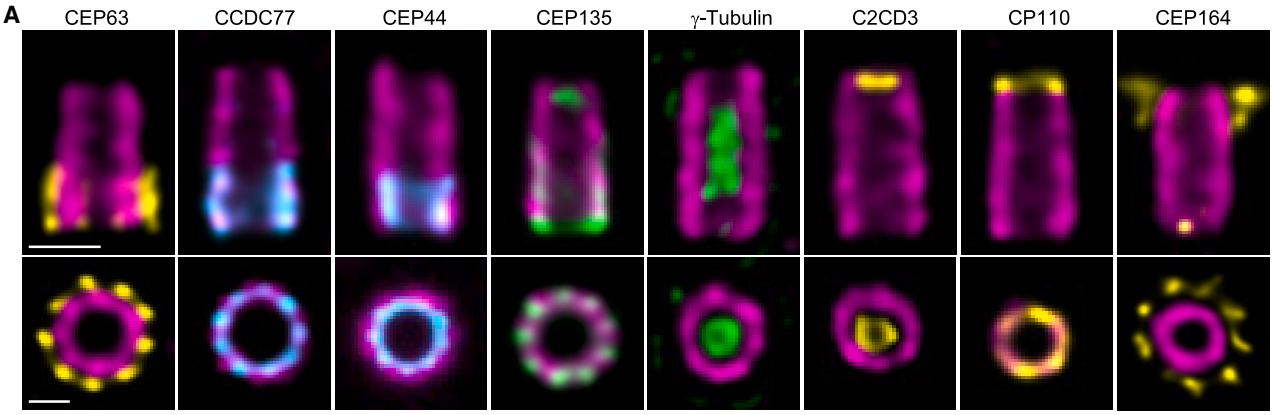
INTRODUCTION

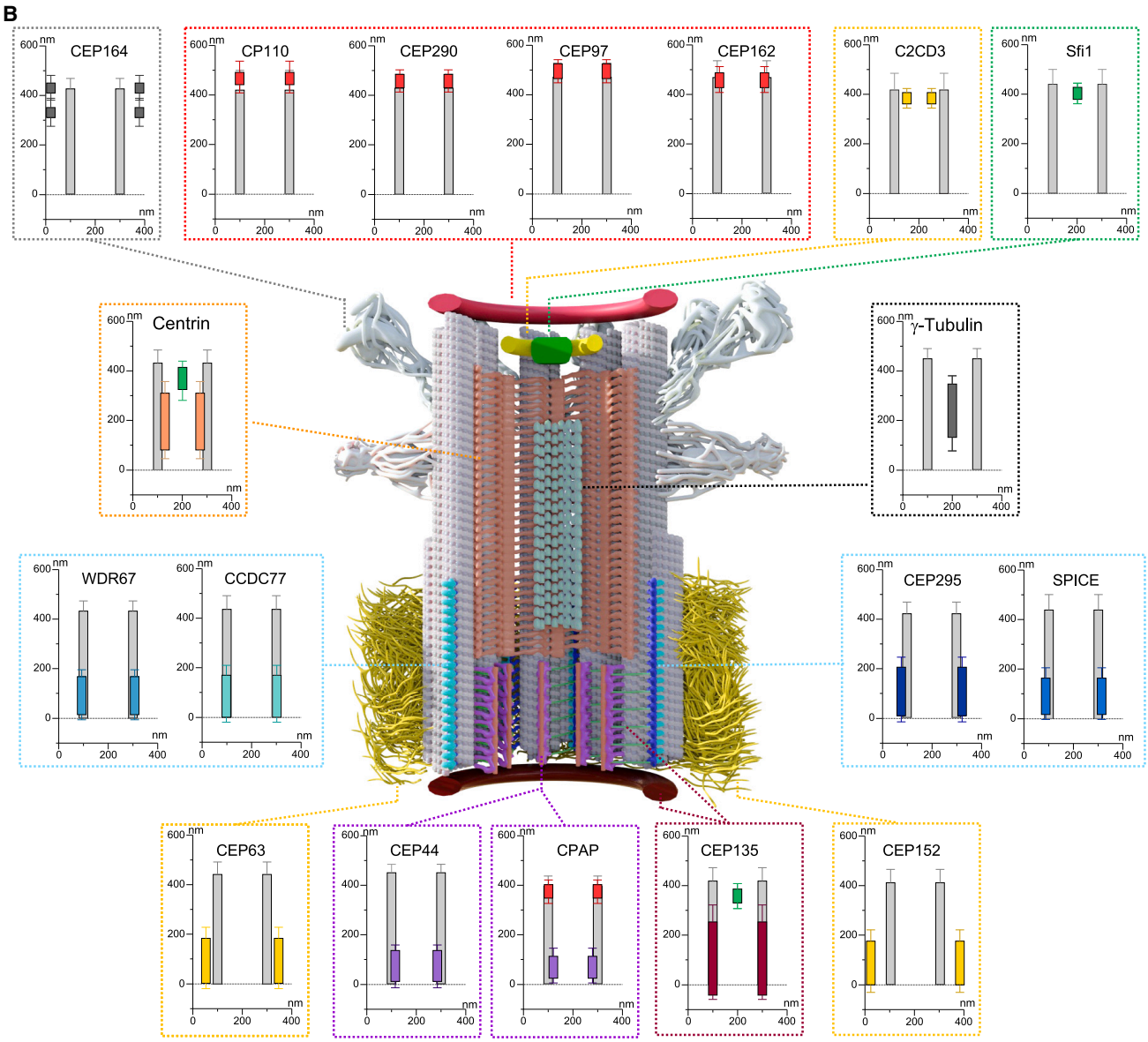
Centriole duplication is a key process essential to ensuring faithful cell division as well as cilia formation.[1,2] Mature mammalian centrioles are large macromolecular complexes of about 2 gigadaltons, characterized by a 9-fold organization of microtubule triplets (MTTs), creating a cylindrical shape about 400–450 nm long and 200–250 nm wide.[3,4] Along the proximal to distal axis, the centriole displays a polarity with three distinct regions, dubbed proximal, central, and distal.[4] The proximal region accommodates three main elements: the cartwheel, consisting of a central hub from which emanate nine radial spokes terminating by a globular region called the D2 density[5,6]; the pinhead, which connects the cartwheel to the MTT[7,8]; and the A-C linker, bridging adjacent MTTs together.[8,9] In the central core region lies the inner scaffold (IS), a large structure that covers up to 70% of the centriole, connects adjacent MTTs/microtubule doublets (MTDs), and is important for maintaining centriole integrity.[10] Finally, the distal end is characterized by the distal and sub-distal appendages located at the external surface, both of which are critical for driving centriole docking to the plasma membrane and ciliogenesis.[11,12]

During the cell cycle, centrioles duplicate once and only once,[13] with the assembly of a procentriole (PC) at the proximal base of each of the two mature centrioles. During this highly controlled process, which starts at the G1/S transition, the various structural elements in the PC must be perfectly assembled to support the function of the mature centriole in the subsequent cell cycle. Forward and reverse genetic screens as well as mass spectrometry analyses, together with cell biology studies in different species, have shed light on the role of many proteins involved in this process.[14–23] However, attributing specific proteins to architectural features within the centriole still represents a major fundamental challenge owing to the resolution limit in optical microscopy, restricting our understanding of their exact structural contribution during biogenesis.

In this study, using super-resolution ultrastructure expansion microscopy (U-ExM), we performed a molecular mapping of the centriole and PC that allows the assigning of centriolar components to each of their structural elements. By combining this approach with a time-series reconstruction method as previously performed for the endocytic process,[24] we reveal the different stages in the formation of the various structural elements of the human centriole. Altogether, these data allow us to establish a quantitative and mechanistic model of the human centriole assembly.

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RESULTS

Correlating molecular localization to structural sub-elements of the mature centriole architecture

To unveil the molecular architecture of the human centriole, we correlate the position of centriolar proteins within specific structural elements (Figure S1) using U-ExM, which allows a nanoscale molecular mapping of proteins.[25] We analyzed the distribution of 21 centriolar proteins in human osteosarcoma U2OS cells (Figures 1 and S2; Data S1). Analysis of the distribution in the radial and longitudinal dimensions of each fluorescent signal was done relative to the α- and β-tubulin signals as a proxy for the centriolar microtubule wall (Figures 1 and S2; Data S1). Note that we found that the acetylated-tubulin marker was not suitable for this purpose, as its signal is lacking at the distal tip of the centriole (Data S1). Based on this nanoscale mapping, proteins were grouped according to their location close to structural elements within the centriole, highlighting possible functional and/or architectural modules.

Starting from the proximal region, we visualized the external torus made of the two pericentriolar scaffold proteins CEP152 and CEP63[26–28] (Figure 1). Consistently, CEP63 was found 47 nm distant from the tubulin wall, while the more external CEP152 was found 80 nm away from the tubulin wall. Longitudinally, both protein signals covered 40% of the centriole length, displaying an average height of 180 nm (Figures 1, S2, and S3A; Data S1). In addition, as recently observed,[29] we found that CEP63 was organized in a 9-fold symmetry with a 40° angle between each density around the centriole (Data S1).

In the same region, the A-C linker structure, spanning 35%–45% of the centriolar length,[10] bridges neighboring MTTs. Up to now, only POC1 has been associated with this architectural element, but this localization remains a subject of debate in various organisms.[10,30–34] We identified four candidates within the A-C linker region: CEP295, important for centriole-to-centrosome conversion as well as MTT and A-C linker formation[31,35]; SPICE, which cooperates with CEP120 and CPAP in centriole elongation[36–38]; and the evolutionary conserved centriolar proteins WDR67 and CCDC77[39] (Figures 1, S2, and S3B; Data S1). Our analysis focused on assessing their structural involvement in the A-C linker based on three main features: (1) longitudinal localization, (2) radial position relative to the microtubule wall, and (3) localization between MTTs. We found that all the four proteins covered between 35% and 45% of the centriole length (Figures S3C and S3D). Radially, CCDC77 and WDR67 were closely colocalized with the tubulin wall, while SPICE and CEP295 were positioned 15 and 22 nm outside of the centriole, respectively (Figures S3E and S3F). This suggests that these four proteins may either be components of the A-C linker or closely associated with it. In this study, we focused on CCDC77 and WDR67, as they were juxtaposed to MTTs. Notably, in centrioles analyzed from the top view, both proteins were located between MTTs, confirming their association with the A-C linker (Figures S3G and S3H). Interestingly, we observed additional localizations for CCDC77 and WDR67 unrelated to the A-C linker, with a weak signal of WDR67 at the torus level comparable in diameter with CEP152 (Figure S3I; Data S1) and CCDC77 present at the distal end of one of the two mature centrioles. Using ciliating RPE1 cells, we demonstrated that this additional localization was present on the mother centriole at the base of the cilium (Figures S3J and S3K, white arrowheads).

The proximal region also entails the triplet base, the pinhead, and the D2 regions connecting the cartwheel (the latter being absent from mature human centrioles) to the MTTs.[4] As expected from the literature, proposing a pinhead localization for CPAP,[6,40] we found that CPAP is located 20 nm internal to the centriolar microtubule wall and covering about 85 nm of the centriole in this region (Figures 1, S2, and S3L; Data S1). Remarkably, we uncovered an additional localization for CPAP, capping centriolar microtubules at the distal end (Figures 1, S2, and S3P; Data S1). This localization corresponds with the role of CPAP as a microtubule-plus-end-binding protein,[41–44] providing the visual explanation for how CPAP can modulate centriolar microtubule elongation. We also found that CEP44, important for the formation of a bona fide centriole wall,[31] localizes in the pinhead region. As CEP44 is a microtubule-binding protein, this localization strongly suggests that CEP44 could be part of one of the pinhead densities, referred to as the pinfoot[8] (Figures 1, S2, and S3L; Data S1). We next analyzed CEP135, previously localized to the cartwheel/pinhead region.[45] We found that CEP135 displays a complex distribution in U-ExM.[46] CEP135 spans the entire centriolar length, exhibiting a variable radius along the proximal to distal axis (Figures 1 and S2; Data S1). In the proximal region, CEP135 protrudes from the microtubule wall with the same diameter as tubulin, while it displays a slightly larger radius in the central region. In the distal region, CEP135 signal is detected in the luminal space, similarly to the localization of SFI1 and centrin, as previously published (Figures 1, S2, and S3Q; Data S1).[10,47]

At the level of the centriole's core region, in addition to the known IS proteins POC1B, POC5, centrin, and FAM161A[10] (Figures 1, S2, and S3M; Data S1), we observed that γ-TUB forms a tube into the lumen of the centriole (Figures 1, S2, S3N, and S3O; Data S1). This distribution, in addition to the known pericentriolar localization (Figures S3N and S3O, arrowheads), is consistent with what has been reported previously.[48,49] Quantification reveals a cylindrical structure 200-nm long and 60-nm in diameter (Figure S2; Data S1), suggesting γ-tubulin's presence as an

Figure 1. Nanoscale mapping of 21 proteins in human mature centrioles

(A) Centrioles observed in side view (top) or top view (bottom). Cells were stained for tubulin (magenta) and CEP63, CCDC77, CEP44, CEP135, γ-tubulin, C2CD3, CP110, and CEP164. Scale bars: 200 (top) and 100 (bottom) nm.

(B) Model of a human mature centriole displaying the structural elements of the centriole along its proximal to distal axis. MTT, gray; pinhead, magenta; triplet base, green chartreuse; A-C linker, cyan; torus, dark yellow; inner scaffold, orange peach; luminal tube, light green; luminal distal ring, green/yellow; distal cap, red; appendages, white. The radial and longitudinal positions of each protein along the centriole are depicted next to the model. Error bars denote SD. See Table S2 for statistics.

See also Figures S1, S2, and S3.

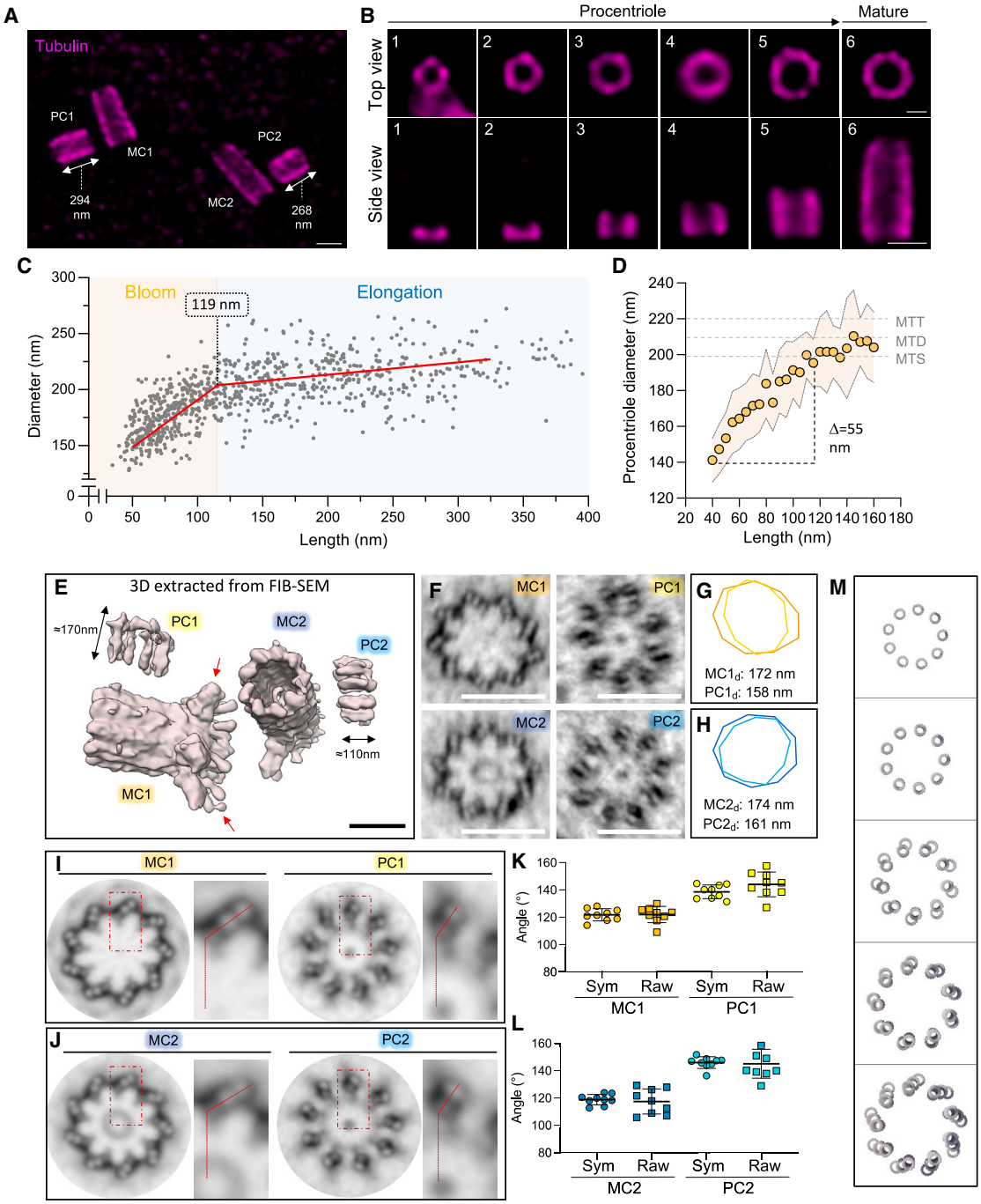


Figure 2. Monitoring the growth of the centriolar microtubule wall

(A) Centrosomes from duplicating cell stained for tubulin. Note the asymmetry of procentriole length (PC1 > PC2 of 26 nm). MC, mature centriole; PC, procentriole. Scale bars: 200 nm.

(B) Centrioles stained for tubulin observed in top view (top) and in side view (bottom). Scale bars: 100 (top) and 200 (bottom) nm.

(C) Evolution of the tubulin diameter over length during procentriole to mature centriole assembly. Measurements of 829 centrioles from >10 independent experiments were fitted with a segmented linear regression (see Figure S4G).

(D) Evolution of the tubulin diameter over length during the early stages of procentriole assembly. Gray dashed lines indicate the simulated diameters of mature centrioles with microtubule singlets (MTSs), microtubule doublets (MTDs), and microtubule triplets (MTTs) based on cryoelectron microscopy (cryo-EM) image (see Figure S4H).

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unidentified structure within the centriole lumen rather than part of the IS.

Distally, our analysis identified two subclasses of proteins. The first includes CP110, CEP290, CEP97, CPAP, and CEP162, known for their function in centriole length and ciliogenesis regulation.[50–52] Quantification reveals that these proteins form a ring-like structure, aligning with the microtubule diameter and capping the centriole end (Figures 1, S2, and S3P; Data S1). The second group comprises CEP135, centrin, and SFI1—appearing as a luminal dot—and the protein C2CD3 (Figures 1, S2, and S3Q; Data S1), which forms an 83-nm internal ring-like structure reminiscent of the recently observed distal ring structure *in situ* with cryo-electron tomography (cryo-ET)[53,54] and previously described as a distal fiber structure.[55] Lastly, we mapped CEP164, a well-known distal appendage protein[20,56] (Figures 1 and S2), confirming its 9-fold symmetry and distinct dual distribution (Figures 1 and S2; Data S1), consistent with previous observations.[56]

Monitoring centriole tubulin wall biogenesis

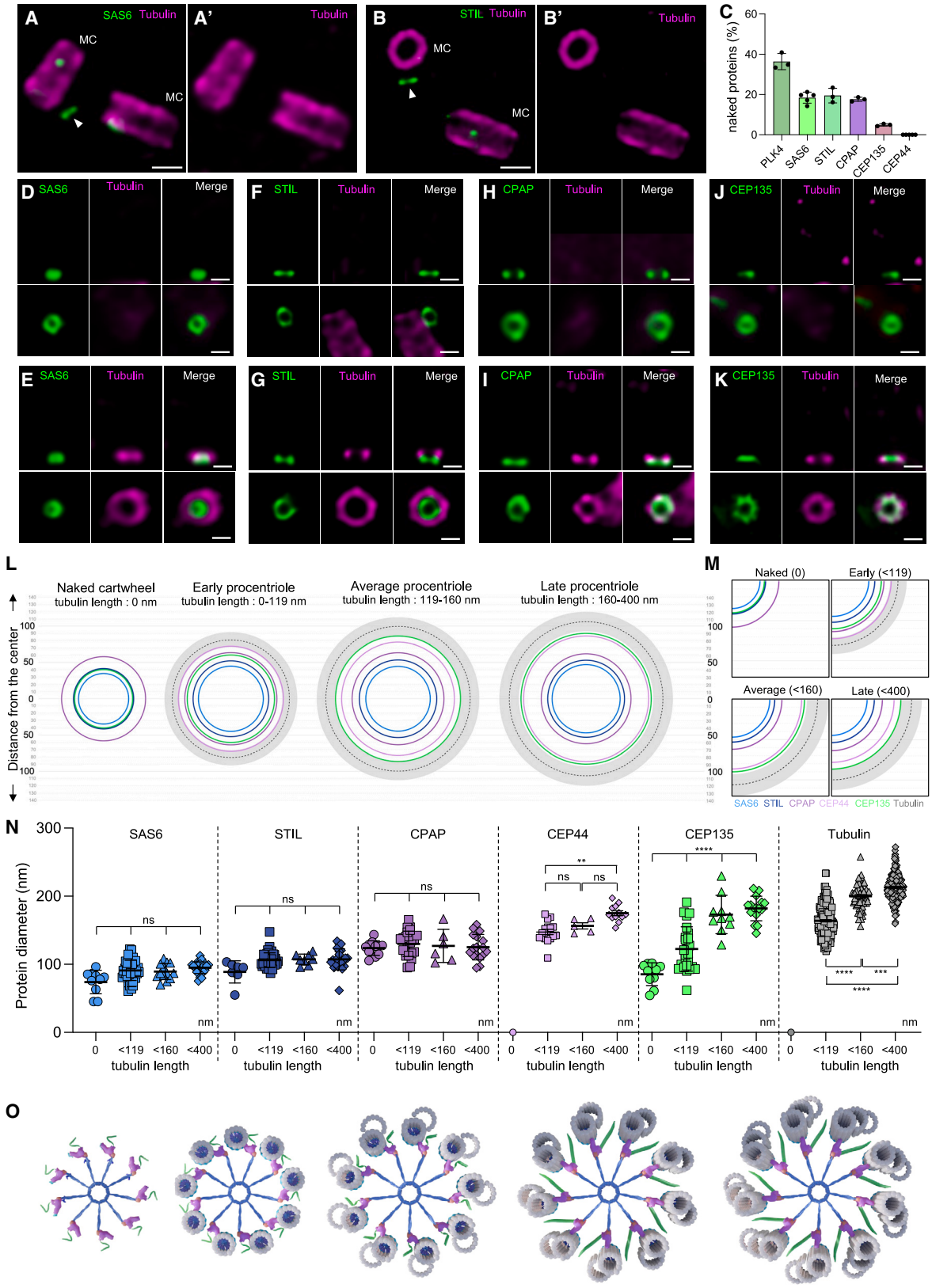
We next investigated PC growth, first using the tubulin signal as a readout (Figures 2A and 2B). Interestingly, regardless of the cell stage, we found that PC growth was asymmetric within a centrosome, with PC derived from the CEP164-positive mature centriole statistically 30 nm longer than the one derived from the daughter centriole (Figures 2A and S4A–S4C). Moreover, analyzing centriole size distribution across the different phases of the cell cycle reveals a non-uniform pattern (Figures S4D–S4F). Therefore, we decided to classify the growth of the PC according to the tubulin length, regardless of the cell-cycle stage, and perform a time-series reconstruction aimed at revealing the centriole assembly process (Figure 2B).

By measuring tubulin length and proximal diameter of over 800 PCs and mature centrioles, we unveiled a bimodal assembly: a "bloom phase" marked by radial and longitudinal expansion until the tubulin length reaches 119 nm, followed by the "elongation phase," whereby the diameter stabilizes while length continues to grow (Figures 2B, 2C, and S4G). One explanation for the increase in diameter could be found in the successive addition of the A-, B-, and C-microtubules during assembly. However, by modeling the U-ExM fluorescent signal of a mature centriole containing either microtubule singlets (MTSs), MTDs, or MTTs, we found that the proximal diameter would only differ by 20 nm between the singlet and the triplet configurations (Figure S4H). By contrast, we quantified the bloom phase as being characterized by an increase of about 55 nm, suggesting that the observed increase may be due to a structural enlargement of the PC (Figure 2D). Interestingly, this tubulin diameter increase has been previously proposed as a mechanism of the iris diaphragm (Figure S4K) during basal body formation in the monkey oviduct.[57,58] Furthermore, ε-tubulin and δ-tubulin mutant centrioles that fail to mature and assemble their MTTs also show a reduction in internal diameter of the centriole, suggesting that this bloom phase precedes full MTT assembly.[59] To validate our findings, we analyzed the three-dimensional (3D) ultrastructure of a human centrosome using focused ion beam milling technology available from Openorganelle.janelia.org[60] (Figures 2E–2L, S4I, and S4J). We extracted a pair of centrioles composed of two MCs (MC1 and MC2) and their two adjacent PCs (PC1 and PC2) (Figures 2E, 2F, S4I, and S4J). By measuring the mean proximal diameter of a nine-sided polygon passing through each A-microtubule, we confirmed that PCs exhibit smaller internal diameters than mature centrioles (Figures 2F–2H and S4I). Moreover, the angle of the MTD/MTT formed with respect to the center of the centriole changes from 140° to 120° between the bloom phase and the mature centrioles (Figures 2I–2L). This difference, previously observed in other systems,[58] suggests a progressive reorientation of microtubule blades during the bloom[61] (Figure 2M; Video S1). We generated a structural model recapitulating microtubule assembly (Figure 2M), first with the formation of the A-, then B-, and C-microtubules, concomitantly with a progressive distancing of the microtubule blades during the bloom phase and a closure of the MTT angle of about 20°.

The naked cartwheel assembly defines the onset of centriole biogenesis

To better understand the bloom phase, we focused next on the cartwheel formation, a crucial structure comprised of 4–6 layers, each layer made of spokes merging every 17–25 nm and connecting the nine microtubule blades (Figure S1).[6,8,62–64] We monitored the two major cartwheel components, SAS-6 and STIL, and the putative pinhead/D2/triplet base proteins CPAP, CEP44, and CEP135, as well as PLK4, a Polo-like kinase crucial for cartwheel assembly.[65–68] Surprisingly, we discovered that, except for CEP44, these proteins were detectable at the site of PC assembly before the microtubules in 36% of all centrosome for PLK4, about 18% for SAS-6/STIL/CPAP, and only 5% for CEP135 (Figures 3A–3K and S5). Moreover, through the super-resolution provided by our U-ExM approach, we clearly detected that these proteins were already forming ring-like structures (Figures 3D, 3F, 3H, and 3J; Data S1) of 77, 75, 89, 120, and 85 nm in diameter, respectively, for PLK4, SAS-6, STIL, CPAP, and CEP135. These observations indicate that PLK4, SAS-6, STIL, and CPAP are probably forming the first cartwheel layer prior to CEP135 and microtubule recruitment, which we will refer

(E) 3D representation of a HeLa centrosome focus ion beam - scanning electron microscopy (FIB-SEM) image (from https://openorganelle.janelia.org/datasets/jrc_hela-2). Note the asymmetry of procentriole length (PC1 > PC2 of 60 nm). Red arrows indicate distal appendages of the mature centriole (MC1). Scale bars: 200 nm.
(F) Top views of the mature centrioles and procentrioles. Scale bars: 200 nm.
(G and H) Diameter representation from MC1/PC1 (G, orange/yellow) and MC2/PC2 (H, dark blue/light blue) measured from images in (F).
(I and J) Circularized and symmetrized images from (F).
(K and L) MTT/MTD angles measured from raw (F) and symmetrized (I and J) images from MC1/PC1 (K) and MC2/PC2 (L). See Table S2 for statistics.
(M) Model of the procentriole bloom with the sequential appearance of the A-, B-, and C-microtubules.
See also Figure S4.



(legend on next page)

to here as a naked cartwheel layer. To ascertain that this result was not due to microtubule destabilization during the fixation process,[69] we analyzed SAS-6 in cryo-fixed and expanded cells (cryo-ExM)[70] and confirmed that SAS-6 rings were detectable before centriolar microtubule assembly (Figures S5E and S5F).

In exploring the subsequent assembly steps during the bloom phase, we predicted that the cartwheel or pinhead regions would enlarge as they directly interact with microtubules via the pinhead. To test this, we measured the diameter of the cartwheel and pinhead components across four pseudo-temporal groups: the first with the naked cartwheel phase (microtubules length: 0 nm), the second with microtubules up to 119 nm in height, the third with microtubules from 119 to 160 nm, and the last one with microtubules from 160 to 400 nm (Figures 3L–3N). Notably, we found that throughout the assembly process, solely CEP135 showed a dramatic change in diameter (from 85 to 182 nm), appearing first as a disk, then as a star-like structure extending between the microtubules, suggesting that the core of the cartwheel does not structurally rearrange radially and that only the outermost part is structurally reorganized (Figures 3K–3N; Data S1).

Modeling the formation of the PC in the bloom phase (Figure 3O), we noted that the pinhead—and the triplet base—must be modified to accommodate the bloom phase. This model reinforces our recent hypothesis that CEP135 could be part of the triplet base,[6] extending during the bloom phase to accommodate the tubulin diameter growth (Figure 3O; Video S2).

The elongation of the cartwheel-containing region

The cartwheel structure in the PC consists of several layers connected to the MTT by the pinhead, and with one or several naked layer protruding from the proximal end.[4–6,71,72] To investigate cartwheel elongation, we next examined the longitudinal positions of key cartwheel proteins, SAS-6, STIL, CEP44, CPAP, PLK4, and CEP135, as well as γ-tubulin, a component of the microtubule-nucleating γ-tubulin ring complex (γ-TuRC) found at the minus end of the A-microtubule during centriole biogenesis in cryo-ET[73] (Figures 4A and 4B). Interestingly, by taking PCs between 120 and 200 nm, i.e., post-bloom phase, we found that SAS-6 and STIL were similarly localized, with an average length of 90 nm of which about 20 nm sticks out from the microtubule end[6] (Figures 4C–4E and S6A). Similarly, we found γ-tubulin strictly below the minus end of the microtubule, also in agreement with the position of the γ-TuRC.[73] CPAP, which is 96 nm in length, mainly locates inside the proximal PC (Figure 1) but also protrudes about 15 nm (Figures 4C–4E and S6A). These observations indicate that SAS-6, STIL, γ-tubulin, and CPAP are components of the naked cartwheel (Figures 4C–4E). Furthermore, cryo-ET studies have indicated that the naked cartwheel lacks the pinhead.[6] Thus, together with CPAP's ability to form elongated fibrils through its G-box domain,[40] which resembles the extended D2 density,[6] we propose that CPAP is part of the D2 density that links the pinhead to the tip of the cartwheel spoke, extending up to the protruding segment.

With regard to CEP135 protruding from the proximal end of mature centriole (Figure 1), we found that it does not extend beyond the tubulin end in the PC, suggesting its absence in the naked cartwheel (Figures 4C–4E and S6A; Data S1) and rather a pinhead/triplet base position.[6] Similarly, the longitudinal localization of CEP44, spanning 78 nm of the PC, correlates with the pinhead position, potentially composing the pinfoot (Figures 4C–4E and S6A; Data S1).

Finally, we mapped PLK4 and found that it localized at the level of the protruding cartwheel, organized as a ring-like structure (Figures 4C–4E and S6A; Data S1). Because PLK4 interacts with STIL,[65] we propose that a fraction of PLK4 is stably associated with the outer side of the cartwheel, capping its proximal end.

To study protein evolution during PC growth, we focused on the distal elongation of SAS-6, STIL, CEP44, CPAP, and CEP135 (Figures 4F–4J; Data S1). We identified three clusters (Figure 4K). Both STIL and SAS-6 display rapid growth before tubulin length reaches about 85 nm (Figure S6) and then slows down drastically (Figures 4F, 4G, 4K, and 4L). CEP44 and CPAP present very similar curves, with progressive growth until the tubulin length reaches 190 nm (Figure S6) and then plateaus (Figures 4H, 4I, 4K, and 4L), while CEP135 appears to elongate continuously with tubulin, independently from the others (Figures 4J, 4K, 4L, and S6). We coupled these data with a structural model (Figure 4M; Video S2) depicting the initial assembly steps from the naked cartwheel, followed by distal elongation of the cartwheel, D2, pinhead, and triplet base at different speeds relative to tubulin.

The MTT connectors

The mature centriole comprises the A-C linker in the proximal region and the IS at the core/distal level,[10] essential for maintaining microtubule cohesion.[46] To understand their assembly

Figure 3. Steps of procentriole initiation

(A–B′) Centrosomes stained for tubulin and SAS-6 (A and A′) or STIL (B and B′). Arrowheads indicate the presence of SAS-6 (A) and STIL (B) in absence of tubulin signal (A′ and B′), thereafter called "naked." MC, mature centriole. Scale bars: 200 nm.

(C) Percentage of cells presenting a signal for PLK4, SAS-6, STIL, CPAP, CEP135, and CEP44 at the level of the future procentriole in the absence of a tubulin signal. See Table S2 for statistics.

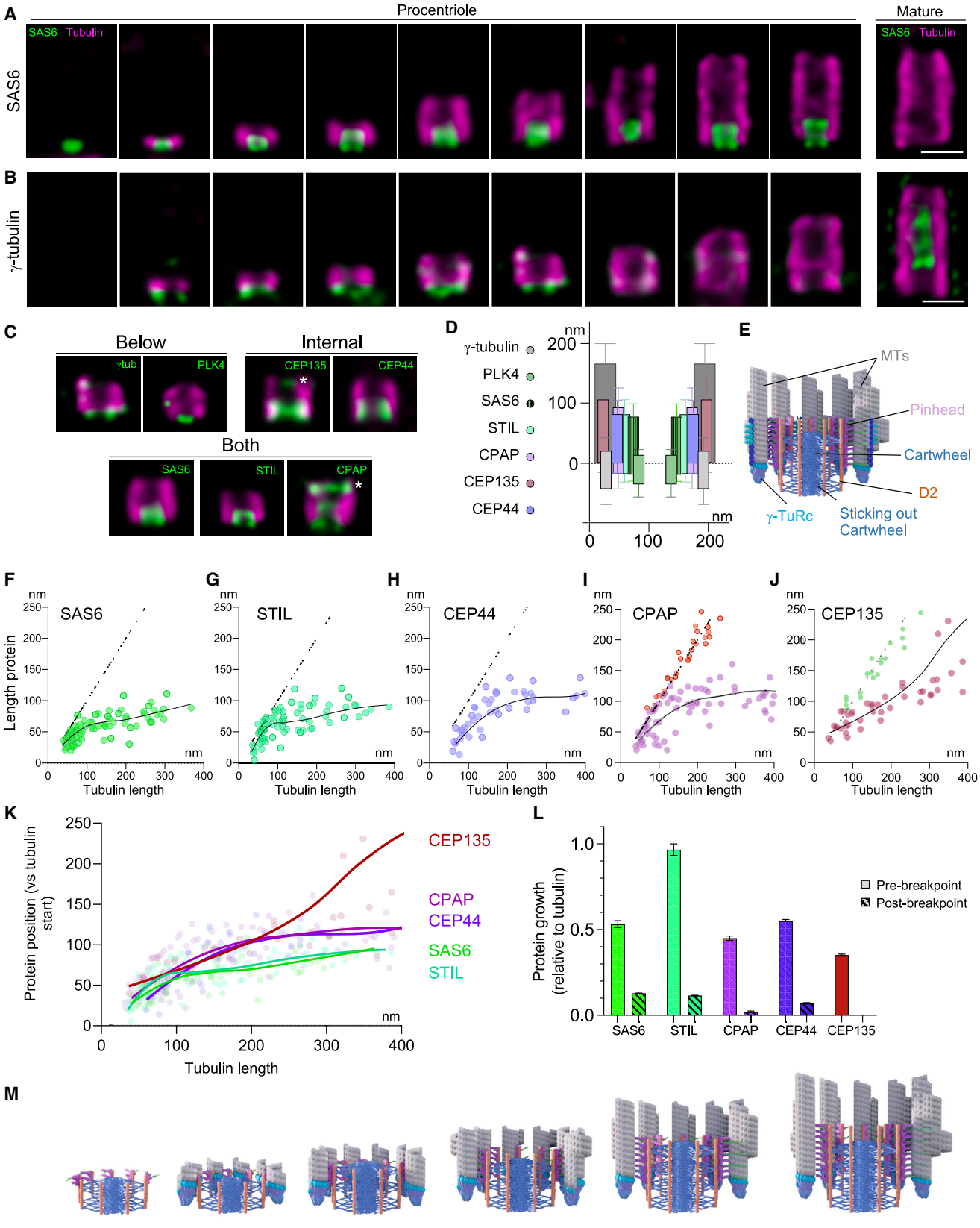
(D–K) Procentrioles stained for tubulin and SAS-6 (D and E), STIL (F and G), CPAP (H and I), and CEP135 (J and K). (D), (F), (H), and (J) show naked SAS-6, STIL, CPAP, and CEP135, and (E), (G), (I), and (K) show the protein in early procentrioles. Scale bars: 100 nm.

(L and M) Normalized diameter representation of SAS-6 (light blue), STIL (dark blue), CPAP (purple), CEP44 (mauve), and CEP135 (green) at different stages of procentriole assembly according to the tubulin (gray) length (0, <119, <160, and <400 nm). All diameters were normalized on the average tubulin diameter. See Table S2 for detailed statistics.

(N) Raw diameters of SAS-6, STIL, CPAP, CEP44, and CEP135 at the same stages of procentriole assembly as in (M). See Table S2 for statistics.

(O) Model of the structural reorganization of the cartwheel during the bloom phase. Initially, the cartwheel is formed with SAS-6 and STIL, and part of the pinhead and D2 density (CPAP). A-microtubules appear with the triplet base (CEP135). The microtubule blades separate from each other, reorganizing the pinhead and triplet base until the procentriole is completely formed before the end of the bloom phase.

See also Figure S5.



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dynamics, we examined the appearance and elongation of A-C linker candidates (CCDC77, WDR67, CEP295, and SPICE) and IS components (POC5, FAM161A, and centrin) during PC formation. We excluded the IS component POC1B,[10] as we could not detect it clearly during PC assembly (Figures S7A and S7B).

Time-series reconstruction revealed two distinct assembly patterns for A-C linker candidates. CEP295 and SPICE appear at the onset of microtubule assembly, while CCDC77 and WDR67 emerge later (Figures 5A–5J). These results suggest a two-stage assembly, probably starting from the A-microtubule linker named A-link.[8] Interestingly, CCDC77 and WDR67 recruitment correlated with the transition from the bloom phase to the elongation phase at 119 nm (Figures 5I and 5J; Data S1), suggesting that their appearance signifies completion of the bloom phase and A-C linker connection between neighboring MTTs. In support of this observation, we also noted that the A-C linker was not visible on the young PCs (Figures 2I and 2J). However, despite a different appearance, all A-C linker candidates exhibited similar unidirectional growth toward the distal part of the centriole, reaching a final size of about 200 nm (Figures 5E–5H and S7C–S7F; Video S2).

Regarding the IS, POC5, FAM161A, and centrin displayed a delayed appearance compared with the A-C linker (Figures 5K–5S). POC5 and FAM161A, known to localize to the centriole in the G2 phase of the cell cycle,[74] were detected at centrioles when the tubulin signal length reaches about 160 nm (Figures 5K, 5L, 5N, and 5O). Consistently, centrin, previously described with a dual localization at both the distal and central part of the centriole,[10] was only detected as a distal dot at the earliest time point of PC growth and appears gradually at the level of the IS when tubulin reaches 160 nm (Figures 5M and 5P). Like the A-C linker, the IS structure does not appear all at once but also grows as the centriole lengthens. We found that the IS grows as fast as tubulin, maintaining a constant distance of 50 nm from the distal end of microtubules and 100 nm from the proximal end (Figures 5Q–5S, S7G, and S7H; Video S2).

Two distinct protein complexes define the distal end from the onset of centriole biogenesis

Centriole length regulation involved distal proteins such as CPAP, C2CD3, CP110, and CEP97,[41–43,50,75,76] with the CP110/CEP97 complex proposed to inhibit elongation.[50] Conversely, CPAP overexpression induces long microtubule structures,[41,42] while its depletion reduces centriole microtubule length.[36,77] Whether these proteins constitute a unique distal module and whether their recruitment is concomitant with tubulin during PC assembly is not yet known.

We monitored the recruitment of nine proteins localizing at the distal end of the centriole during assembly, namely CEP290, CEP97, CP110, CEP162, CPAP, SFI1, C2CD3, centrin, and CEP135 (Figures 6 and S8). As previously described for the centrin distal dot and SFI1,[47] all the distal proteins were detected at the onset of PC assembly, concomitantly with the microtubule wall (Figure 6; Data S1). However, we uncovered two distinct localization patterns. CP110, CEP97, CEP290, CEP162, and CPAP are capping the centriole microtubule distal end, as a ring-like structure being positioned on top of the tubulin signal throughout growth (Figures 6A–6C; Data S1). By contrast, SFI1, centrin, CEP135, and C2CD3 initially localized at the centriole distal end but gradually moved internally during assembly, appearing below the microtubule distal end in the final stage (Figures 6D–6F and S8J). Consistently, these proteins do not localize at the microtubule level but rather more in the distal lumen, with SFI1, centrin, and CEP135 localizing as dots and C2CD3 forming a ring-like structure of 83 nm (Figure 6D; Data S1). These distinct behaviors and localizations suggest that the two groups of proteins form two distal complexes: a distal cap (CP110, CEP97, CEP290, CEP162, and CPAP) and the luminal distal ring (SFI1, centrin, CEP135, and C2CD3) (Figure 6G).

Unveiling the architectural choreography of the centriole assembly

Utilizing a time-series reconstruction approach, we investigated the assembly of centriolar sub-elements via their associated proteins. We wondered whether the studied proteins could be grouped into clusters that could be averaged to better reflect the overall behavior of the assembly of centriolar sub-elements. To unambiguously unveil the level of similarity of individual centriolar proteins, we performed a multidimensional scaling analysis (MDS) on 20 different proteins (Figure 7A). We obtained six clusters aligned with specific structural components: SAS-6/STIL (cartwheel), CPAPp/CEP44 (pinhead), CCDC77/WDR67/SPICE/CEP295 (A-C linker), FAM161A/POC5 (IS), CEP135d/C2CD3/SFI1/centrind

Figure 4. Procentriole elongation

(A and B) Centrioles during assembly and at mature stage stained for tubulin (A) and (B) and SAS-6 (A) or γ-tubulin (B). Scale bars: 200 nm.

(C) "Average" procentrioles (180–200 nm in length) stained for tubulin and γ-tubulin, PLK4, CEP135, CEP44, SAS-6, STIL, and CPAP. Asterisks indicate additional distal localization of CPAP and CEP135.

(D) Relative protein longitudinal and radial positions compared with tubulin. See Table S2 for statistics.

(E) Structural model of the procentriole, lateral view.

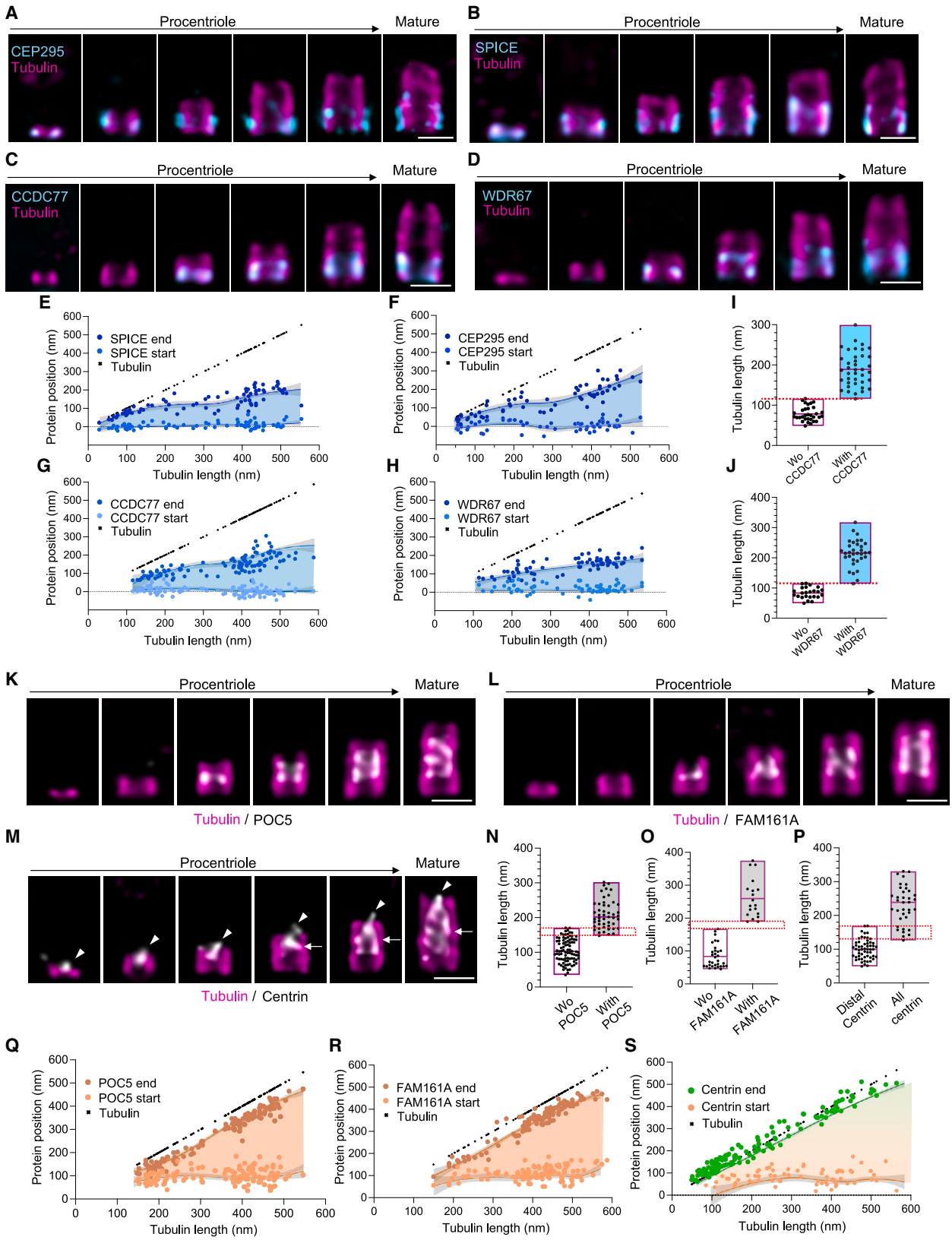
(F–J) Evolution of SAS-6 (F), STIL (G), CEP44 (H), CPAP (I), and CEP135 (J) length over tubulin growth during procentriole assembly. CPAP and CEP135 present a dual localization in the proximal (CPAP purple and CEP135 dark red) and distal (CPAP red and CEP135 green) regions. This figure focuses on the proximal localization (see Figure 6 for distal analysis). The evolution of each set of data was represented as loess curves (black line) to estimate the behavior of each protein during their growth (relative to tubulin).

(K) Merged data from graphs (F) to (J) with the dot plots and the loess curves for each protein, showing three main clusters based on their similar behaviors (SAS-6/STIL, CPAP/CEP44, and CEP135). The breakpoints, when the growth rates change significantly, are indicated in Figure S6.

(L) Protein growth rate relative to tubulin growth before (plain bars) and after (striped bars) breakpoint. See Table S2 for statistics.

(M) Model of the formation and elongation of the cartwheel and procentriole during the bloom phase. A cartwheel layer assembles before γ-TuRC recruitment and microtubule blade assembly. The microtubules elongate approximately two times faster than the cartwheel structures, the pinhead, the D2 density, and the triplet base. During assembly, a cartwheel layer is always lower than the microtubules. After about 100 nm of cartwheel elongation, it stops growing, in contrast to the pinhead and D2 density, which grows to about 190 nm.

See also Figure S6.



(legend on next page)

(luminal distal ring), and CP110/CEP97/CEP290 (distal cap). Notably, CPAPd/CEP162 initially associated with the microtubule cap and joined the luminal ring, indicating potential molecular interplay. CEP135p seemed to behave independently of the other clusters, probably reflecting a more complex localization spanning different structural elements.

Plotting the average assembly behavior of these clusters against tubulin growth (Figures 7B and 7C) revealed two major trends: a linear growth for the distal complexes and IS and a slower growth followed by distinct inflection points for the proximal complexes (cartwheel, pinhead, and A-C linker) (Figures 7C and 7D). The cartwheel and pinhead exhibited growth at 0.75 and 0.5 relative to tubulin, respectively, plateauing at 80 and 162 nm. The A-C linker grew at half the speed of tubulin until 138 nm, slowing down to 0.3 of tubulin speed.

Integrating these findings, we proposed a comprehensive model of centriole biogenesis (Video S3). Initiation involves a naked cartwheel layer with D2 density, followed by the A-microtubule capped by the γ-TuRC, and pinhead formation with the appearance of a distal cap and luminal distal ring. The bloom phase, concomitantly with B- and C-microtubule addition, corresponds to PC diameter increase. Upon reaching 119 nm, the elongation phase begins, together with the completion of the A-C linker, and at 160 nm, the IS assembles. Throughout, the cartwheel protrudes about 20 nm from the proximal end. Interestingly, in mature centrioles, CPAP shortens (Data S1) while CEP135 extends proximally (Data S1). These observations suggest that the loss of the cartwheel has resulted in a trimming process of the D2 density, allowing CEP135 to extend proximally, presumably forming the platform that can recruit the centrosome linker protein C-NAP1 between the two mature centrioles.[78,79] Once the centriole is fully formed, the torus, the γ-tubulin tube, and appendages appear in the G1 phase of the subsequent cell cycle (Data S1).

To conclude, our U-ExM time-series reconstruction paves the way for revealing organelle biogenesis by correlating nanoscale protein localization with sub-structural elements across assembly stages.

DISCUSSION

In this study, we dissected the choreography of the structural assembly of the human centriole and established principles common to other model systems.

60 years ago, electron microscopy studies in *Paramecium* and *Tetrahymena* revealed that the cartwheel can be detected before A-microtubule formation, which led to the model in which the cartwheel imposes the 9-fold symmetry of the centriole.[62,80] Here, by labeling the cartwheel components SAS-6 and STIL, we were able to reveal by U-ExM that the cartwheel devoid of microtubules, called the naked cartwheel, is also present in about 20% of human centrosomes. During the bloom phase, the naked cartwheel persists, protruding proximally, as observed in the human centriole—*Naegleria*, *Chlamydomonas*, and *Paramecium*—using cryo-ET.[6] The function of the naked cartwheel is not known. It may be used to recruit and maintain the γ-TuRC at the base of A-microtubules, but so far, γ-TuRC has only been observed in human centrosomes.[73]

The time-series analysis of the cartwheel assembly enabled the visualization of cartwheel growth. Interestingly, research on *Chlamydomonas* has indicated that the cartwheel undergoes substantial elongation in the PC before being trimmed, resulting in a smaller cartwheel in the parent centriole.[71] This implies that a long cartwheel may be necessary for assembly but is no longer needed in the mature centriole. In our investigation, cartwheel trimming was not observed, as it undergoes degradation during human mitosis.[63] However, we did observe the concurrent disappearance of PLK4 and the γ-tubulin below the MTTs, likely coinciding with cartwheel removal. Surprisingly, CPAP, which protrudes alongside the naked cartwheel, seems to be trimmed, retaining only its microtubule localization (Data S1). By contrast, CEP135, which is not present in the naked cartwheel, starts to protrude a few dozen nanometers below the centriole post cartwheel (Data S1). This implies that the trimming mechanism may be conserved and play a pivotal role in centriole formation.

We also discovered a phase termed the bloom phase, characterized by the enlargement of the centriolar wall as it grows. Unlike the iris diaphragm model of the Albrecht-Buehler model,[57] our model indicated that the A-microtubules are separated from each other and not in contact, such as in *Paramecium*, where A-microtubules are distant by 20–40 nm.[62] Moreover, in Albrecht-Buehler's model, the MTT angles are fixed (Figure S4K), while our FIB-milling analysis confirmed Stubbelfield and Brinkley's observation showing that the MTT angles change between PC and the mature centriole.[81] The mechanism behind this reorientation remains unclear. However, it may involve the A-C linker, as this structure can twist in an iris-diaphragm-like motion in the

Figure 5. The MTT connectors assembly

(A–D) Procentrioles during assembly and mature centriole stained for tubulin and CEP295 (A), SPICE (B), CCDC77 (C), and WDR67 (D). Scale bars: 200 nm.

(E–H) Evolution of the start and end position of SPICE (E), CEP295 (F), CCDC77 (G), and WDR67 (H) signals relative to tubulin growth during procentriole assembly. The growth of each protein was represented as loess curves (blue lines). The light blue region depicts the region of the centriole that is covered by the protein signal.

(I and J) Centriole length, based on the tubulin signal, with and without (Wo) CCDC77 (I) or WDR67 (J). The dotted red line represents the average tubulin length when the CCDC77 and WDR67 signal appears (≈115 nm). See Table S2 for statistics.

(K–M) Procentrioles during assembly and mature centriole stained for tubulin and POC5 (K), FAM161A (L), and centrin (M). Arrowheads and arrows indicate the luminal distal localization and the microtubule-associated central localization of centrin, respectively (M). Scale bars: 200 nm.

(N–P) Centriole length with and without (Wo) POC5 (N), FAM161A (O), and central centrin (P). The dotted red square represents the average tubulin length when POC5, FAM161A, and central centrin signals appear (≈160, 170, and 150 nm, respectively). See Table S2 for statistics.

(Q–S) Evolution of the start and end position of POC5 (Q), FAM161A (R), and centrin (S) signals relative to tubulin growth during procentriole assembly. The growth of each protein was represented as loess curves (green/orange lines). The peach/peach-to-green regions depict the region of the centriole that is covered by POC5, FAM161A, or centrin signals.

See also Figure S7.

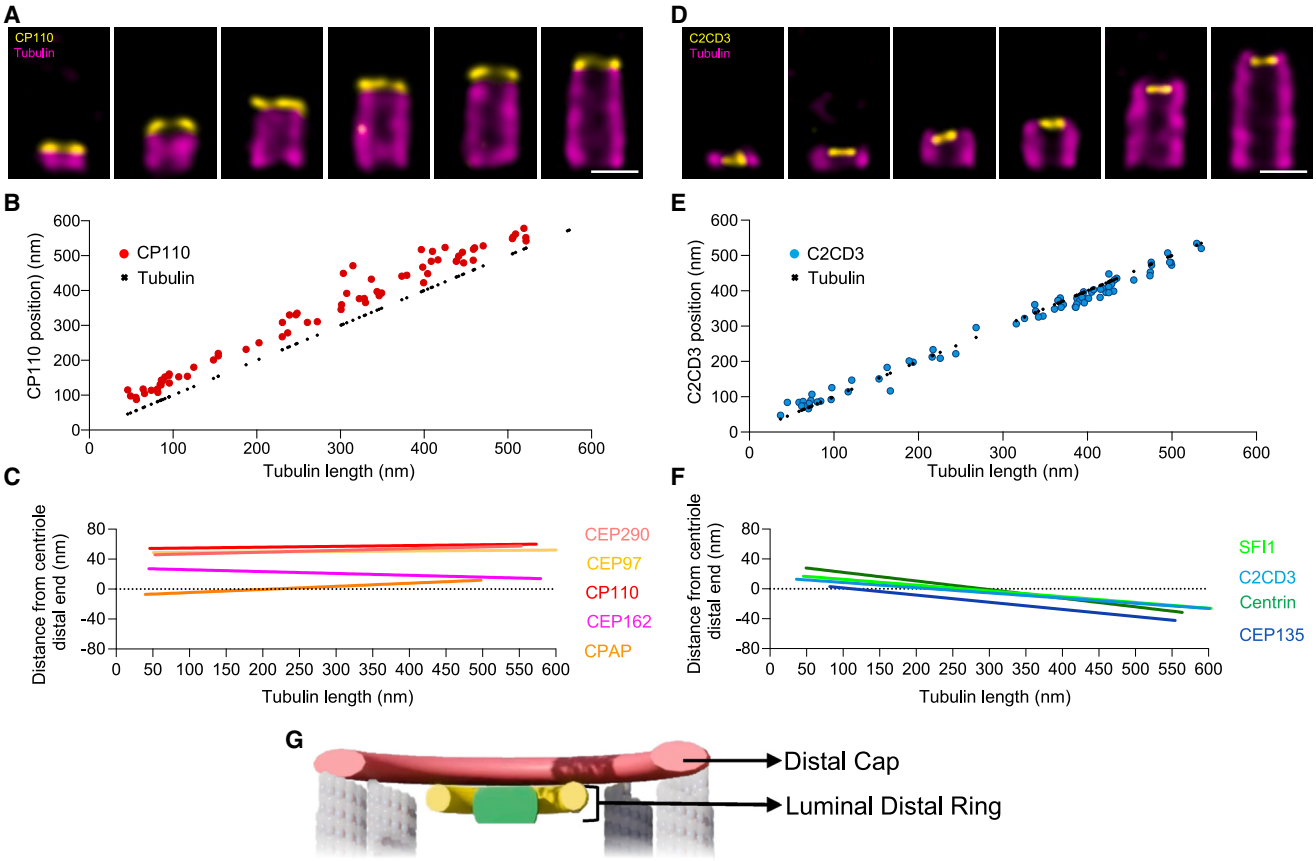


Figure 6. Two distal complex rings

(A) Procentrioles during assembly and mature centriole stained for tubulin and CP110. Scale bars: 200 nm.

(B) Evolution of CP110 position signal relative to tubulin growth during procentriole assembly.

(C) Linear regression curves showing the position of the distal proteins CEP290, CEP97, CP110, CEP162, and CPAP-distal, relative to the end of the tubulin signal during centriole assembly.

(D) Procentrioles during assembly and mature centriole stained for tubulin and C2CD3. Scale bars: 200 nm.

(E) Evolution of position of C2CD3 signal relative to tubulin growth during procentriole assembly.

(F) Linear regression curves showing the position of the distal proteins SFI1, C2CD3, centrin-distal, and CEP135-distal relative to the end of the tubulin signal during centriole assembly.

(G) Model of the structures present at the distal level of the centriole, the distal cap (red) that comprises proteins that localize on the plus end of microtubule triplets, and the luminal distal ring (green and yellow).

See also Figure S8.

longitudinal direction of the PC.[30] Moreover, CEP135 appears to be one of the proteins involved in this process—a protein known to be vital for the assembly of centrioles/basal bodies in various species.[82–84]

During the assembly of the PC, MTTs are connected by a structure known as the A-C linker, first described in 1960.[9] Here, we identified four likely components, namely CEP295, SPICE, CCDC77, and WDR67. Interestingly, we observed two-phase behavior: CEP295 and SPICE were detected from the early stages of microtubule assembly, while CCDC77 and WDR67 appeared toward the end of the bloom phase. This disparity likely reflects the sequential assembly of each microtubule within the MTT complex. It should be noted that our study did not investigate the specific function of the A-C linker at the centriole, but it is hypothesized that it plays a role in maintaining the structural integrity of the proximal region. Now that the constituent proteins have been identified, future experiments will be conducted to test this hypothesis.

In our mapping, we found CPAP localized at both proximal and distal cap regions, an observation in line with the fact that CPAP interacts with STIL[85,86] but also with the distal component CP110.[44] Interestingly, similar to CP110, which blocks microtubule depolymerization and decreases their growth,[44,50,76] CPAP was also characterized as a protein that slows microtubule growth and prevents catastrophes.[43,44] Similarly, CEP135 exhibits a dual distribution pattern, both at the proximal/central region and within the luminal distal ring. How are these dual locations regulated? One explanation involves the potential existence of isoforms. Interestingly, it has been demonstrated that a truncated form of CEP135, lacking the CPAP- and

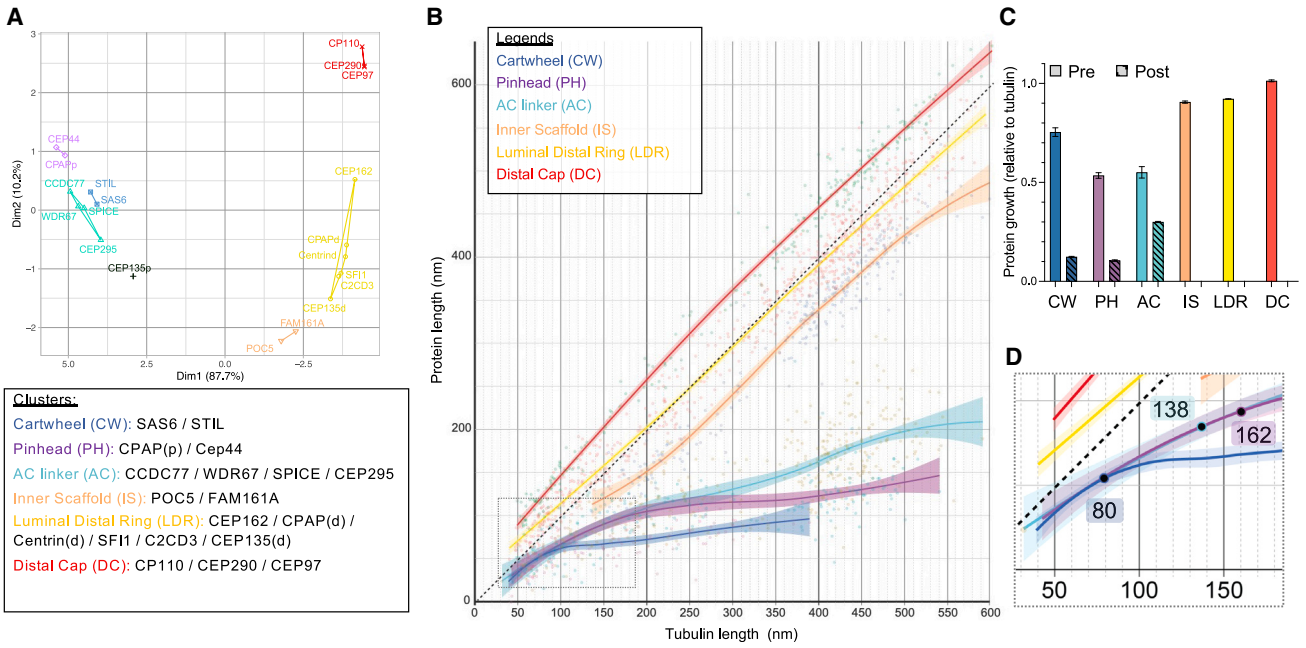


Figure 7. Choreography of the molecular architecture of centriole biogenesis

(A) Clustering of the 20 proteins exhibiting a quantifiable growth during centriole assembly based on their average growth curves' distance. The x axis separates growth based on their linearity from nonlinear (left) to near-linear growth (right), while the y axis describes the amount of deviation. The clusters formed by statistical analyses matched with centriolar structural elements, named as follows: cartwheel (SAS-6 and STIL), pinhead (CPAPp and CEP44), A-C linker (CCDC77, WDR67, SPICE, and CEP295), inner scaffold (POC5 and FAM161A), luminal distal ring (CEP135p, SFI1, centrin, C2CD3, CPAPd, and CEP162), and distal cap (CEP97, CEP290, and CP110). Note the individual behavior of CEP135p, which could not be associated with any clusters.

(B) Fits of loess regressions with confidence bands for the clusters in (A). The dotted black line corresponds to the tubulin, exhibiting an exact linear growth (slope = 1). The graph highlights the clear separation between the nonlinear growth clusters (cartwheel, pinhead, and A-C linker) and near-linear growth clusters (inner scaffold, luminal distal ring, and distal cap).

(C) Chart graph showing the average growth rate for each cluster prior to and after the breakpoints calculated from loess curves (B).

(D) Zoom in of graph (B) highlighting the breakpoints of the clusters with nonlinear growth (see Figures S6 and S7 for details).

SAS-6-binding domains, named CEP135 mini, is involved in regulating centriole assembly.[87,88] This divergence could grant CEP135 the ability to be localized distally. In the future, employing super-resolution techniques to examine the spatial distribution of various protein isoforms within centrioles will undoubtedly enhance our comprehension of their respective functions.

Overall, the reconstruction of centriole architecture assembly presented here not only deepens our understanding of centriole formation but also paves the way for a more nuanced examination of protein functions within this intricate structure. The principles established through this study offer valuable insights that can be applied to diverse model systems, ultimately contributing to a broader comprehension of centriole biology. As we unravel the complexities of centriole assembly, we open doors to new avenues of research that may hold the key to unlocking broader implications in cellular organization and function.

Limitations of the study

Our linear model for centriole assembly may not fully capture the dynamic nuances. Notably, microtubules could undergo growth and shrinkage, akin to cytoplasmic microtubules. Additionally, microtubules might experience phases of accelerated growth and pauses. Using our data to calculate an average growth rate of microtubules, we found that it may vary across cell-cycle phases (Figures S4D–S4F). Moreover, analyzing centriole size distribution across phases reveals a non-uniform pattern, indicating fast growth at the beginning followed by a slowdown. However, questioning whether these data truly mirror microtubule dynamic assembly is pertinent. Additionally, understanding the dynamic behavior of centriole sub-element proteins adds complexity. To address these uncertainties, live imaging of protein loading, as previously undertaken,[33,89] is imperative. Yet, the challenge lies in the limited resolution, making it arduous to differentiate between localization at the centrosome and incorporation at the structural level. Future investigations, utilizing advanced techniques like super-resolution live imaging or correlative live and expansion microscopy, hold promise in overcoming limitations and providing insights into unanswered questions.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.cell.2024.03.025.

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AUTHOR CONTRIBUTIONS

M.H.L. performed and analyzed all the experiments in the paper with the help of D.G. É.B. and V.L. performed the U-ExM on distal proteins and CEP152, respectively. L.B. supported M.H.L. on the A-C linker candidates. S.B. provided technical help. J.M.N. did the statistical analysis of the clusters and the breakpoints. V.H. and P.G. designed the project and wrote the manuscript with input from all authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal anti-alpha-tubulin	Abcam	Cat#ab18251 RRID:AB_2210057
Mouse monoclonal alpha-tubulin	ABCD antibodies	Cat# AA345
Mouse monoclonal beta-tubulin	ABCD antibodies	Cat# AA344
Mouse monoclonal acetylated-tubulin	ThermoFisher	Cat# 32-2700 RRID:AB_2533073
Mouse monoclonal anti-gamma-tubulin	Abcam	*Cat# ab11316 RRID:AB_297920*
Rabbit polyclonal anti-gamma-tubulin	Gift from A-M Tassin	N/A
Mouse monoclonal anti-centrin	Merk Millipore	*Cat# 04-1624 RRID:AB_10563501*
Rabbit polyclonal anti-C2CD3	Atlas antibodies	*Cat# HPA038552 RRID:AB_10669542*
Rabbit polyclonal anti-CCDC77	Proteintech	*Cat# 26369-1-AP RRID:AB_2880495*
Rabbit polyclonal anti-CEP44	Proteintech	Cat# 24457-1-AP, RRID:AB_2879557
Rabbit polyclonal anti-CEP63	Proteintech	Cat# 16268-1-AP, RRID:AB_2077079
Rabbit polyclonal anti-CEP63	Millipore	Cat# 06-1292, RRID:AB_10918481
Rabbit polyclonal anti-CEP97	Proteintech	Cat# 22050-1-AP, RRID:AB_11182378
Rabbit polyclonal anti-CEP135	Proteintech	Cat# 24428-1-AP, RRID:AB_2879543
Rabbit polyclonal anti-CEP152	Thermo Fisher Scientific	Cat# A302-479A, RRID:AB_1966085
Rabbit polyclonal anti-CEP162	Atlas Antibodies	Cat# HPA030173, RRID:AB_10602206
Rabbit polyclonal anti-CEP164	Proteintech	Cat# 22227-1-AP, RRID:AB_2651175
Rabbit polyclonal anti-CEP290	Thermo Fisher Scientific	Cat# PA5-84641, RRID:AB_2791792
Rabbit polyclonal anti-CEP295	Atlas Antibodies	Cat# HPA038596, RRID:AB_10672720
Rabbit polyclonal anti-CP110	Proteintech	Cat# 12780-1-AP, RRID:AB_10638480
Rabbit polyclonal anti-CPAP	Proteintech	Cat# 11517-1-AP, RRID:AB_2244605
Rabbit polyclonal anti-FAM161A	Le Guennec et al.[10]	N/A
Rabbit polyclonal anti-PLK4	Proteintech	Cat# 28750-1-AP, RRID:AB_2881206
Rabbit polyclonal anti-POC1B	Thermo Fisher Scientific	Cat# PA5-24495, RRID:AB_2541995
Rabbit polyclonal anti-hPOC5	Thermo Fisher Scientific	Cat# A303-341A, RRID:AB_10971172
Mouse monoclonal anti-SAS6	Santa Cruz Biotechnology	Cat# sc-81431, RRID:AB_1128357
Rabbit polyclonal anti-SFI1	Proteintech	Cat# 13550-1-AP, RRID:AB_10596938
Rabbit polyclonal anti-SFI1	Laporte et al.[47]	N/A.
Rabbit polyclonal anti-SPICE	Thermo Fisher Scientific	Cat# A303-272A, RRID:AB_10950135
Rabbit polyclonal anti-STIL	Thermo Fisher Scientific	Cat# A302-441A, RRID:AB_1944268
Rabbit polyclonal anti-WDR67	Atlas Antibodies	Cat# HPA023710, RRID:AB_1858819
Goat polyclonal anti-Mouse IgG-A488	ThermoFisher	A11029
Goat polyclonal anti-Rabbit IgG-A488	ThermoFisher	A11008
Goat polyclonal anti-Mouse IgG-A568	ThermoFisher	A11004
Goat polyclonal anti-Rabbit IgG-A568	ThermoFisher	A11036
Chemicals, peptides, and recombinant proteins		
Acrylamide (AA, 40%)	SIGMA	A4058
Formaldehyde (FA, 36.5–38%)	SIGMA	F8775
Sodium Acrylate (SA, 97–99%)	SIGMA / AK Scientific	408220 / 7446-81-3

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
N, N-methylenbisacrylamide (BIS, 2%)	SIGMA	M1533
Tetramethylethylendiamine (TEMED)	ThermoFisher	17919
Amonium Persulfate (APS)	ThermoFisher	17874
ethane:propane 37%:63%	PanGas	N/A
Acetone (99.8% AcroSeal)	Acros Organics	cat. no. 67-64-1
Nuclease-free water	Ambion-ThermoFisher	AM9937
Poly-D-Lysine	ThermoFisher	A3890401
tetracycline-negative fetal calf serum	ThermoFisher	15393691
DMEM GlutaMAX	ThermoFisher	10566016
Penicillin / Streptomycin	ThermoFisher	15140122
Deposited data		
Focused Ion Beam milling Image of human centrosome	Xu et al.[60]	https://openorganelle.janelia.org/datasets/jrc_hela-2
Experimental models: Cell lines		
hTERT RPE-1 (immortalized Retinal pigmental epithelial cells)	ATCC	CRL-4000
U20S (human osteosarcoma cells)	ATCC	HTB-96
Software and algorithms		
Fiji	Schindelin et al.[90]	https://imagej.net/Fiji
GraphPad Prism	GraphPad Software, LLC	https://www.graphpad.com/
R	R Development Core Team, 2020[91]	www.R-project.org/
UCSF Chimera	Pettersen et al.[92]	https://www.cgl.ucsf.edu/chimera/
CropAndResize plugin	This study	https://zenodo.org/doi/10.5281/zenodo.10802024
PickCentrioleDim plugin	This study	https://zenodo.org/doi/10.5281/zenodo.10802038
CentrioleGraph plugin	This study	https://zenodo.org/doi/10.5281/zenodo.10802046
How-to-use-the-plugins tutorial (video)	This study	https://zenodo.org/doi/10.5281/zenodo.10802048
EM-to-U-ExM-resolution	This study	https://zenodo.org/doi/10.5281/zenodo.10802050
MDS_centriole	This study	https://zenodo.org/doi/10.5281/zenodo.10802058

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Paul Guichard (paul.guichard@unige.ch).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All data reported in this paper will be shared by the lead contact upon request.
- All original code and plugins (with tutorials) have been deposited on GitHub (https://github.com/CentrioleLab) and Zenodo, and are publicly available as of the date of publication. DOIs are listed in the key resources table.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human cell culture

U2OS (human osteosarcoma cells; ATCC-HTB-96) and hTERT RPE1 (Retinal Pigment Epithelial cells; ATCC-CRL-4000) cells were cultured in DMEM medium supplemented with GlutaMAX (Life Technology), 10% tetracycline-negative fetal calf serum (Life Technology), penicillin and streptomycin (100 μg/ml) at 37°C and 5% CO2. Both U2OS and RPE1 cells were seeded at 35000 cell/cm^2 on coverslips as previously published.[47] Cells were regularly tested for mycoplasma contaminations.

METHOD DETAILS

Antibodies used in this study

See key resources table, Table S1, and Data S1 for details regarding the primary antibodies. Secondary fluorescent antibodies were purchased from Invitrogen (A11008, A11004, A11029, and A11036, Invitrogen, ThermoFisher) and used at 1:800 dilutions for standard immunofluorescence experiments and 1:400 for U-ExM.

Ultrastructure Expansion Microscopy (U-ExM)

U-ExM was performed as described previously.[47] Briefly, U2OS and RPE1 were grown on 12 mm coverslips and processed for expansion except for Figure S7A where staining was performed on non-expanded cells. No pre-fixation step was performed except for Figures S5E and S5F where cells were cryo-fixed by manual plunge freezing and freeze substitution[70] to validate that the presence of naked proteins (without tubulin) was not due to tubulin destabilization during the expansion procedure. Briefly, cells on coverslips were rapidly plunged into liquid ethane cooled with liquid nitrogen at -180 °C. Coverslips were then rapidly transferred into nitrogen-chilled acetone, placed on dry ice and agitated overnight to allow the temperature to rise to −80 °C. Samples were further incubated without dry ice for 1.5 h until the temperature reached ∼0 °C. Samples were then rehydrated in successive ethanol:water baths, 5 min each, as follows: ethanol 100%, ethanol 100%, ethanol 95%, ethanol 95%, ethanol 70%, ethanol 50% and PBS and proceed for expansion.

Following cryo-fixation or not, coverslips were incubated in 2% AA + 1.4% FA diluted in PBS for 3 to 5 h at 37°C prior to gelation in monomer solution (19% sodium acrylate, 0.1% bis-acrylamide, and 10% acrylamide) supplemented with TEMED and APS (final concentration of 0.5%) for 30 min to 1 h at 37°C. Denaturation was performed for 1 h 30 min at 95°C and gels were stained as follows. After 2x20 min of washes in ddH_2O allowing expansion, gels were measured with a caliper. The gel expansion factor was obtained by dividing the size after expansion by 12 mm, which corresponds to the size of the coverslips use for sample seeding. Measurements of lengths and diameters were scaled according to the expansion factor of each gel. Gels were placed in PBS for 15 min and incubated with primary antibodies in PBS-BSA 2% for 2h30 at 37°C or overnight at room temperature with agigation. After 3 washes in PBS-tween 0.1%, gels were incubated with secondary antibodies in PBS-BSA 2% for 2h at 37°C with incubation and washed 3 times in PBS-tween 0.1%. Gels were processed for final expansion by 3 incubations of 20 min in ddH_2O.

The following reagents were used in U-ExM experiments: ethane:propane 37%:63% (PanGas), acetone (99.8% AcroSeal, cat. no. 67-64-1, Acros Organics), formaldehyde (FA, 36.5–38%, F8775, SIGMA), acrylamide (AA, 40%, A4058, SIGMA), N, N-methylenbisacrylamide (BIS, 2%, M1533, SIGMA), sodium acrylate (SA, 97–99%, 408220, SIGMA and 7446-81-3, AK Scientific), ammonium persulfate (APS, 17874, ThermoFisher), tetramethylethylendiamine (TEMED, 17919, ThermoFisher), nuclease-free water (AM9937, Ambion-ThermoFisher), and poly-D- lysine (A3890401, Gibco).

Imaging

Expanded gels were mounted onto 24 mm coverslips coated with poly-D-lysine (0.1 mg/ml) and imaged with an inverted confocal Leica TCS SP8 microscope. Images were taken with a 63× 1.4 NA oil objective with lightning mode at max resolution, adaptive as "Strategy" and water as "Mounting medium" to generate deconvolved images. 3D stacks were acquired with 0.12 μm z-intervals and a 35 nm x, y pixel size. For the Figures S5G and S7A, images were acquired with an inverted widefield Leica DM18 microscope using a 63× 1.4 NA oil immersion objective and the Thunder "Small volume com- putational clearing" mode and water as "mounting medium" to generate deconvolved images. 3D stacks were acquired with 0.21 μm z- intervals and a 100 nm x, y pixel size.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data representations and quantifications

Unless specified, deconvolved images were used for representations and quantifications. To obtain the centriole assembly friezes, a small z-projection (stack of 4 images maximum) was applied to deconvolved the images. Images were duplicated and rotated to orientate each centriole (pro or mature) in the proximal to distal orientation. A crop of 50*75 pixels was made around each centriole and images were sorted by tubulin length (Figures S9A–S9C). For top viewed-oriented centrioles, only single plane images are presented as a crop of 50*50 pixels.

For each experiment, expanded gel is precisely measured with a caliper and the calculated expansion factor is applied in every quantification. Values presented in graphs and scale bars always correspond to "real" values after application of the expansion factor.

Longitudinal measurements from side viewed centrioles

Measurements of the length, the relative position, and the coverage of centriolar proteins were done on dual staining images where tubulin is always used as a proxy for the evaluation of the centriole length. From resized images where the pixel size was artificially decreased by 6 to improve the measurement precision (plugin ''CropAndResize'', https://github.com/CentrioleLab), the fluorescent signal distribution of tubulin and the protein of interest were measured using the Fiji line scan (with a width of 200-220 pixels) and the plot profile tool. Using the plugin ''PickCentrioleDim'' (https://github.com/CentrioleLab) to facilitate the picking of the start and the end of the fluorescent signal (defined as 50% of the peak value at both extremities of the centriole), the user automatically generates an entry table with the coordinates of fluorescent signal extremities in each measured channel. For raw length measurements, the distance between the fluorescent signal extremities was calculated after application of the gel expansion factor and plotted using GraphPad Prism7. For the coverage, the calculation was done as previously indicated and the length of the protein of interest was expressed as a percentage of the total length of the centriole and plotted using GraphPad Prism7. For the relative protein position, the tubulin was defined as the reference protein and its starting coordinate was shifted and set to 0. The same shift was applied to the protein of interest to keep the correct distances between the 2 proteins. For each protein, the average and side deviation of the starting position and of the ending position were calculated and pasted into an csv entry table. The entry table was red by a second plugin (''CentrioleGraph'', https://github.com/CentrioleLab) which generates a plot illustrating the relative average position of the protein of interest to the tubulin. The radial position of the protein relative to tubulin can be loaded in the parameters of the plugin (distance between the protein and the microtubule wall in nm). The first number, indicating 0 by default, corresponds to tubulin position, the second number corresponds to the protein of interest radial position and can be modified according to the radial measurement (see next section).

Note that all plugins and tutorial videos are available on github (https://github.com/CentrioleLab/) and Zenodo (see key resources table).

Radial measurements from top viewed centrioles

Measurements of diameter, radial position, and position relative to microtubule triplets were done on dual staining images where tubulin was always used as a proxy for the evaluation of the centriole diameter. From resized images, where the pixel size was artificially decreased by 6 to improve the measurement precision (CropAndResize plugin, https://github.com/CentrioleLab), the fluorescent signal distribution of tubulin and the protein of interest were measured using the Fiji line scan across the centriole and the plot profile tool. Depending on the protein of interest behavior, either full diameter or shift between the tubulin signal and the protein signal (radial position) was measured. In both cases, using the plugin ''PickCentrioleDim'' (https://github.com/CentrioleLab) to facilitate the picking of the peaks of the fluorescent signal, the user automatically generates an entry table with the coordinates of fluorescent signal extremities in each measured channel. For raw diameter measurements, the distance between the fluorescent signal peaks was calculated for each channel after application of the gel expansion factor and plotted using GraphPad Prism7. For the radial measurements, the distance between the peak signal of tubulin and the peak signal of protein of interest was calculated after application of the gel expansion factor and plotted using GraphPad Prism7. Note that all plugins and tutorial videos are available on github (https://github.com/CentrioleLab/) and Zenodo (see key resources table).

For the position relative to microtubule triplets, only centrioles where all the 9 microtubules triplets could be observed, using the tubulin staining, were selected and a line scan was drawn onto two successive triplets. The plot profile tool was used to save and plot the full fluorescent profile of both tubulin and the protein of interest along this line. Individual plots were aligned on the first triplet peak of fluorescence and average plot profile was represented using GraphPad Prism7.

Normalized diameter

In Figures 3L–3N and S2B, normalized diameters were represented based on measurements described above. To generate the graphs of Figures 3L–3N, the average tubulin diameter of ''average procentriole'' (from 160-400 nm) was used as a reference and normalized to 200 nm. The same normalization factor was applied to all the other values. For Figure S2B, average diameters at the proximal, central and distal regions were measured from 20-25 centrioles stained for tubulin and a marker of each region.

Measurements of diameter versus length during procentriole assembly

To assess the diameter changes during procentriole assembly, measurements of both length and diameter of the proteins of interest were done on side-viewed centrioles (see ''longitudinal measurements from side viewed centrioles''). Briefly, using Fiji Line scan, plot profile tool and the 2 plugins ''CropAndResize'' and ''PickCentrioleDim'', we measured centrioles in both dimensions for each channel (from proximal to distal and from left to right). This gave 4 measurements: lengths of tubulin and protein of interest , diameters of tubulin and protein of interest. Diameters of protein of interest were plotted against the tubulin lengths using GraphPad Prism7 as presented in Figures 2C, 2D, 3N, S4G, and S5D and Data S1.

Note that all plugins and tutorial videos are available on github (https://github.com/CentrioleLab/) and Zenodo (see key resources table).

Procentriole asymmetry

Measurements of procentriole length asymmetry was performed from side views as described in ''longitudinal measurements from side viewed centrioles'' from images of centrosome stained for tubulin and CEP164, a distal appendages marker restricted to mother centriole. Only images where both procentrioles were clearly visible in side views were considered for quantification. The length of the procentriole deriving from the mother centriole (CEP164-positive) was plotted against to its paired procentriole deriving from the daughter centriole (CEP164-negative) using GraphPad Prism7 (Figure S4B). The length difference between each pair of centriole

(ΔProcentriole length) was obtained by subtracting the length of the daughter procentriole to the length of the mother procentriole (Figure S4C).

FIB-SEM data analysis

FIB-SEM data used in Figures 2E–2L, S4I, and S4J were downloaded from OpenOrganelle website: https://openorganelle.janelia.org/datasets/jrc_hela-2, interphase HeLa Cells ID jrc_hela-2. The 3D view was obtained with the UCSF chimera software.[92] The 3D volume was generated using first a gaussian filter of 1 and a surface representation with an adapted threshold of intensity. To remove small densities, the function "hide dust" was used. Top views of the procentrioles were generated using ImageJ software[90] and the ''reslice'' function to reorient the centrioles and obtain an adequate view. Circularized and symmetrized images were generated using the ImageJ plugin CentrioleJ.[8] The centriole diameters were determined through a two-step process. Firstly, manual tracing was conducted using Fiji's "polygon selection" tool to meticulously connect all the A-tubules. Next, automatic measurements in Fiji were performed to obtain the major and minor diameters of the encompassing ellipse. The final value represents the average of these major and minor ellipse diameters, providing an accurate representation of the centriole's diameter. Doublets/Triplets angles were measured using the ''Angle'' tool from Fiji from raw and symmetrized images and plotted using GraphPad Prism7.

Appearance of the MTT connector proteins

The appearance of the MTT connector proteins (CCDC77, WDR67, POC5, FAM161A and centrin) was evaluated by measuring the length of the tubulin fluorescent signal in growing procentrioles. Measurements were sorted depending on whether the fluorescent signal of the MTT connector protein could be detected or not and plotted as two separated set of data. The average between the highest centriole length without fluorescent signal and the lowest centriole length with fluorescent signal was considered as the appearance point of the MTT connectors.

Percentage of naked protein

The quantification of the percentage of cells showing the presence of a centriolar protein in absence of tubulin signal was done manually under a widefield microscope. Briefly, expanded U2OS were stained for tubulin (magenta) and the protein of interest (green). Cells were screened on the tubulin channel to identify G1 cells (two mature centrioles, no procentriole), S/G2 cells (two mature centrioles and at least one procentriole) and mitosis (presence of the mitotic spindle). On each category of cell, the presence or absence of the green fluorescent signal (protein of interest) was counted. Naked proteins were considered when the green fluorescent signal could be detected but not the magenta fluorescent signal (tubulin) at the level of the procentriole. Results were expressed as a percentage of the total amount of cell counted per experiment and plotted using GraphPad Prism7.

Microtubule singlets, doublets and triplets at U-ExM resolution

As previously published,[25] U-ExM resolution images generated in Figure S4H were made from a cryo-EM model of a centriole with microtubule triplets, but this approach can be used with any type of centriole image (as described here: https://github.com/CentrioleLab/EM-to-U-ExM-resolution and Zenodo, see key resources table). First, a scaled image of the centriole is opened with Fiji. In order to mimic the size of the centriole after expansion, the scale of the image is modified using the ''Analyze/Set Scale'' tool and multiplying the value of ''Known distance'' by 4 (expansion factor). The resulting image gave a centriole with a diameter around 1000 nm. Then the contrast is adjusted using the ''process/Binary/Make Binary'' tool. After this step, the microtubules are in black and the background in white. If details other than the triplets are retained, they are manually erased by drawing a rectangle over the identified area followed by the ''Edit/Clear'' tool. The image is then duplicated twice to create 3 images. In the first image, the microtubule triplets are preserved. In the second image, the C-microtubule is manually deleted using the previous approach. In the third image, B- and C- microtubules are manually deleted, leaving just the A microtubule. The images were then filtered at the confocal resolution (≈200nm) using the tool ''Process/FFT/Bandpass Filter''. To do so, we set the "Filter large structures down to" to the maximum, for example 10,000 pixels, and the "Filter small structures up to" to the number of pixels corresponding to 200 nm. To determine this number of pixels, we used the ''Analyze/Set scale'' tool and multiplied the "Distance in pixels" by 200. The contrast of the generated images is finally inverted using the ''Image/Lookup Tables/Invert LUT'' tool. The final images were used to measure the diameters of the three types of centriole, with either triplets, doublets or singlets. The measurements were obtained using ImageJ's plot profile tool, which involved measuring the distances to 50% of the external height of the two Gaussians.

Statistical analysis

Images were analyzed using Fiji.[90] Statistical analyses were perfomed using MS-Excel, GraphPad or R, as indicated below with details and are provided in the figure legends and results. All data are expressed as the mean (average) +/− the standard deviation (SD) or, as usual for regressions, twice the standard errors for 95% confidence intervals. The sample sizes (n) and what they represent (e.g., number of centrioles, number of replicates) are stated in figure legends and/or results. Two-tailed Student's t tests were performed for normally distributed data and are indicated in figure legends and/or results.

Determination of regression models

As the raw data produce uneven point distributions that render the cluterisation difficult, we have softened the raw data with a smoother using the R environment for statistical computing,[91] exploratory plots made using the *ggplot2* graphics system.[93] The growth data for each protein was first screened using regular loess fits (Locally Estimated Scatterplot Smoothing) as this method makes less assumptions about the data than other smoothing methods. Several span values, the parameter that controls the

smoothness of the regression, have been tested in order to assess a possible effect of the data's noise in the fit but these have not shown an observable effect and hence the usual default of 0.75 was used. The smoothed loess regression where used to determine break points (when the growh curve shows a significant change in growth rate). The quality of the loess fits was assessed visually and formally by fitting them with segmented linear model of the protein behaviour (see Figures 4K–4M, 7A–7D, S4G, S6B–S6F, S7C–S7H, and S8A–S8I).

To model the growth of the proteins as a linear model (regression) based on the tubulin length, we used segmented or piece-wise regression with a pre-specified number of break points, based on direct observation of the loess curve and by comparison with the results of the elbow method as implemented in the *inflection* R package.[94,95] These analyses were performed with the *segmented* R package.[96,97]

MDS analysis and clustering of the centriolar proteins

The MDS analysis was conducted to evaluate the behaviour of 19 studied proteins in a non-biological biased manner. The ''distance'' between the protein growth curves was used to establish clusters and to corellate these clusters with known centriolar structural element.

The average ''distance'' between all pairs of proteins was determined based on i) the loess curve behaviour and ii) the spatial localization of the protein relative to tubulin (used as a proxi of centriole growth). The matrix of growth curve distances was then passed to a clustering algorithm (partition around medoids, PAM) and the adjustments for the given numbers of groups (k) were assessed from analysis of the usual statistics (within sums of squares [wss]), silhouette and gap_stat) using the R packages cluster[98] and factoextra[99] (Figures S9D–S9F).

The retained partition, k=7, was then represented using a principal components projection with colour identifying each cluster. A two dimensions plot was found sufficient to depict the unveiled k=7 structure.

The MDS script and data can be found on Github: https://github.com/CentrioleLab/MDS_centriole and Zenodo (see key resources table).

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Supplemental figures

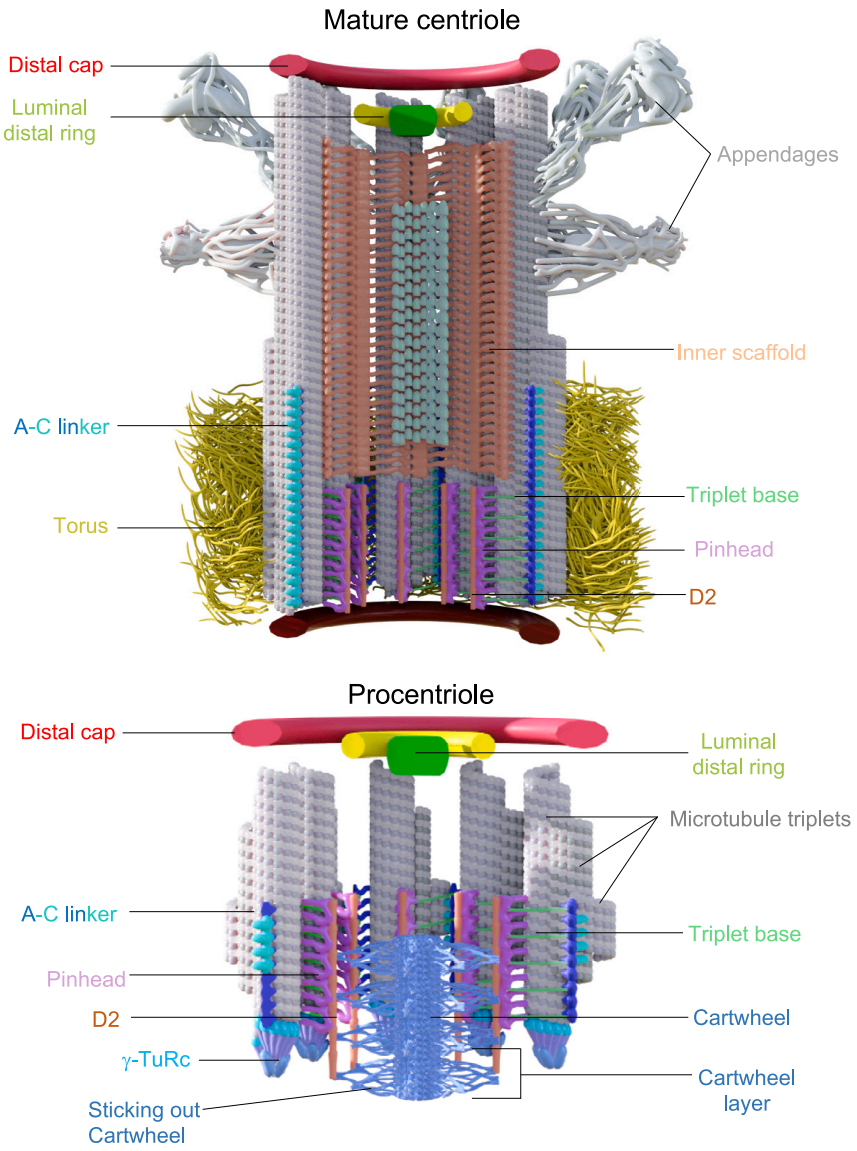


Figure S1. Index of centriole structures, related to Figure 1
Scheme of the procentriole and mature centriole structural elements and their corresponding legends.

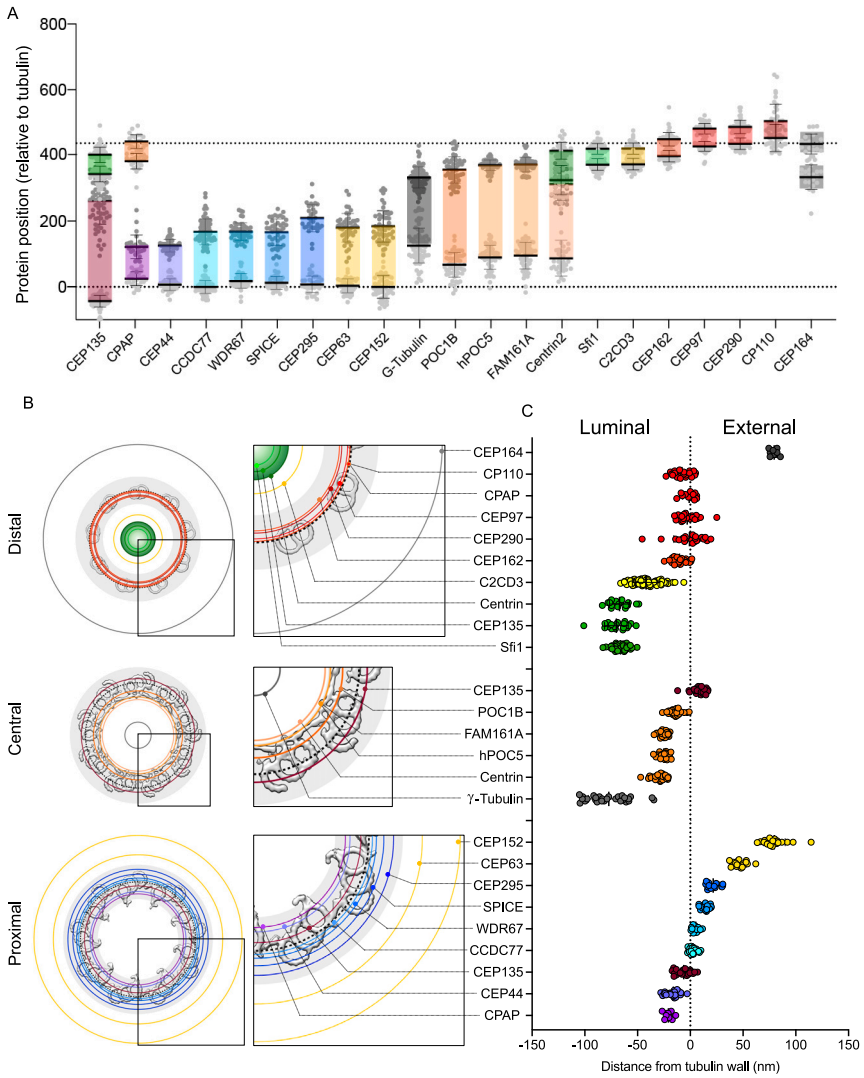
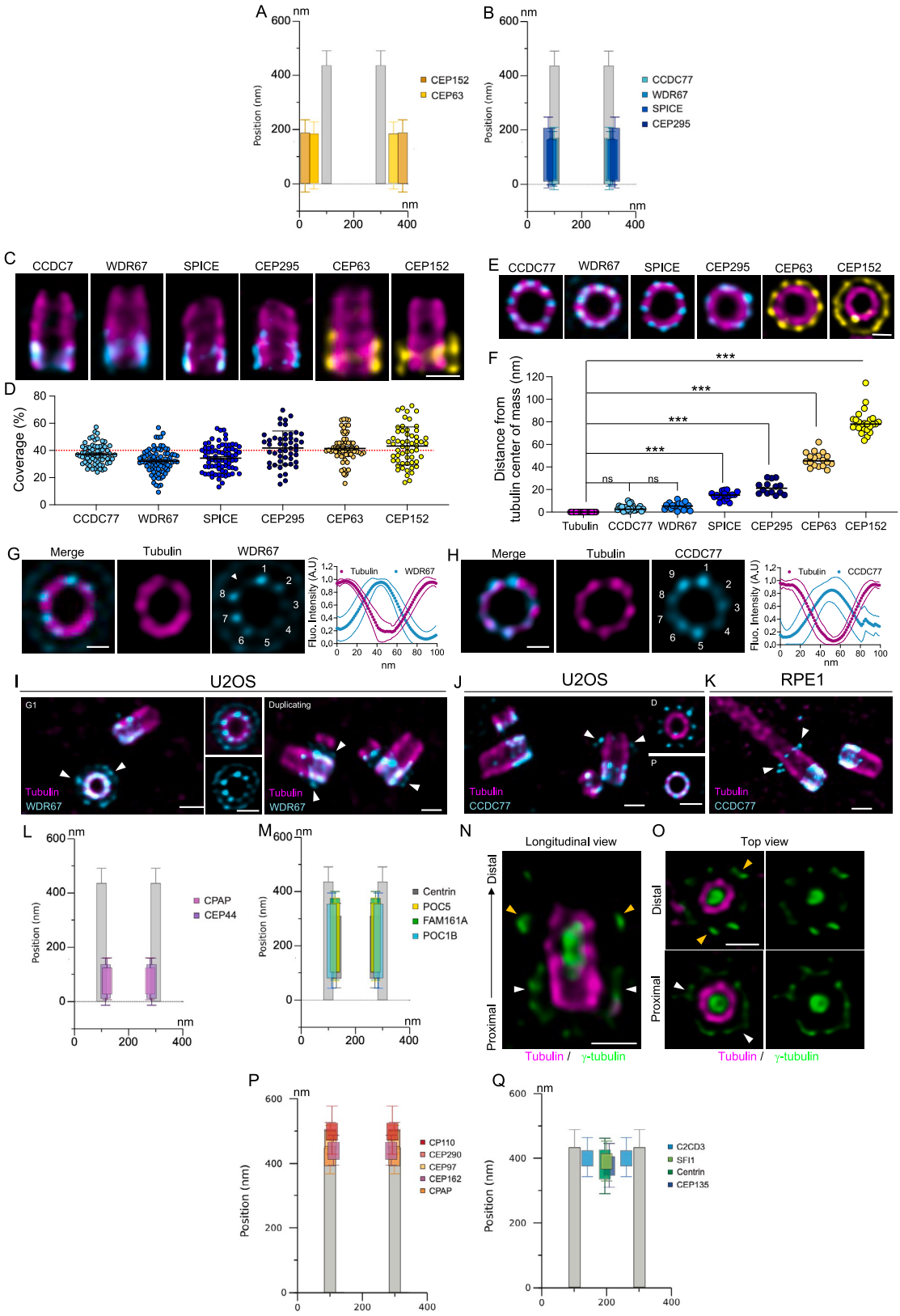


Figure S2. Mapping the human mature centrioles by U-ExM, related to Figure 1

(A) Longitudinal positions of the 21 mapped proteins relative to the normalized tubulin signal. Black dotted lines indicate the start and the end of the tubulin signal, depicting the centriole length. For each protein, the start (light gray dots) and end (dark gray dots) position of the fluorescent signal is presented. Note that for the distal proteins only present as a thin layer, one unique peak of fluorescence could be detected, and, therefore, the measurement reflects the thickness of the fluorescence signal around the peak. Each filled color bar represents the coverage of the indicated protein inside the centriole. See Table S2 for statistics.

(B) Graphical representation of the radial positions of the 21 mapped proteins relative to the tubulin signal, juxtaposed to a cryo-EM map of a *Chlamydomonas* centriole.[10] Black dotted circles indicate the tubulin signal, depicting the MTT wall diameter in the proximal, central, and distal regions. Each color circle corresponds to the diameter of the indicated protein based on measurements presented in (C) and normalized on the average tubulin diameter at each specific region (see STAR Methods). Tubulin diameter (average ± SD): proximal = 197 ± 15 nm; central = 181 ± 13 nm; distal = 167 ± 22 nm. *n* = 26, 20, and 24 centrioles from three independent experiments for proximal, central, and distal, respectively. See Table S2 for statistics.

(C) Distance between the MTT wall (tubulin signal) and the indicated proteins (in nm). A distance below 0 indicates that the protein is localized inside the centriole, while a distance above 0 indicates a protein that is localized outside the centriole. See Table S2 for statistics.



(legend on next page)

Figure S3. Detailed features of protein clusters in human mature centrioles, related to Figure 1

(A) Overlapping radial and longitudinal positions of the torus components, CEP63 (yellow) and CEP152 (mustard), relative to tubulin signal (gray). See Table S2 for statistics.

(B) Overlapping radial and longitudinal positions of the A-C linker candidates CCDC77, WDR67, SPICE, and CEP295 (light blue to dark blue) relative to tubulin signal (gray). See Table S2 for statistics.

(C) Side views of centrioles from U2OS cells stained for tubulin (magenta) and, from left to right, CCDC77, WDR67, SPICE, CEP295, CEP6, and CEP152. Scale bars: 200 nm.

(D) Coverage of indicated protein, expressed as a percentage of the tubulin length. Average ± SD are as follows: CCDC77 = 37.2% ± 7.4%, WDR67 = 32.1% ± 9.0%, SPICE = 34.3% ± 9.7%, CEP295 = 41.8% ± 12.6%, CEP63 = 41.5% ± 10.2%, and CEP152 = 43.3% ± 14.0%. *n* = 70, 81, 76, 48, 70, and 57 centrioles for CCDC77, WDR67, SPICE, CEP295, CEP63, and CEP152, respectively.

(E) Top views of centrioles stained for tubulin (magenta) and, from left to right, CCDC77, WDR67, SPICE, CEP295, CEP63, and CEP152. Scale bars, 100 nm.

(F) Distance between the tubulin signal and the indicated protein signal. Average ± SD are as follows: tubulin = 0 ± 0 nm, CCDC77 = 2.6 ± 3.1 nm, WDR67 = 5.2 ± 3.0 nm, SPICE = 14.5 ± 3.5 nm, CEP295 = 21.7 ± 5.8 nm, CEP63 = 46.7 ± 6.3 nm, and CEP152 = 79.6 ± 10.3 nm. *n* = 28, 18, 19, 14, 18, 19, and 25 centrioles for tubulin, CCDC77, WDR67, SPICE, CEP295, CEP63, and CEP152, respectively.

(G and H) Centrioles from top view stained for tubulin (G and H) and WDR67 (G) or CCDC77 (H). Note that both WDR67 and CCDC77 are located in between each microtubule triplet visualized thanks to tubulin staining. Fluorescence intensity profile along two successive MTTs demonstrating the precise position of WDR67 (G) and CCDC77 (H) are shown on the right of the figures.

(I) Centrosomes stained for tubulin and WDR67 showing a torus-like localization of the protein (white arrowheads), on top of its A-C linker localization. Note that this torus-like pattern is present at only one centriole in G1 and at both once centrioles start to duplicate, reminiscent of the behavior of the torus proteins CEP63 and CEP152. Scale bars: 200 nm.

(J) Centrosomes stained for tubulin and CCDC77 showing an additional localization toward the distal end of the centriole, organized in a 9-fold symmetry outside the tubulin wall (white arrowheads). Note that this additional localization is only present at one of the two mature centrioles. Scale bars: 200 nm.

(K) Centrioles from ciliated RPE1, stained for tubulin and CCDC77, showing the additional localization of CCDC77 at the distal end of the basal body undergoing ciliogenesis (white arrowheads). Scale bars: 200 nm.

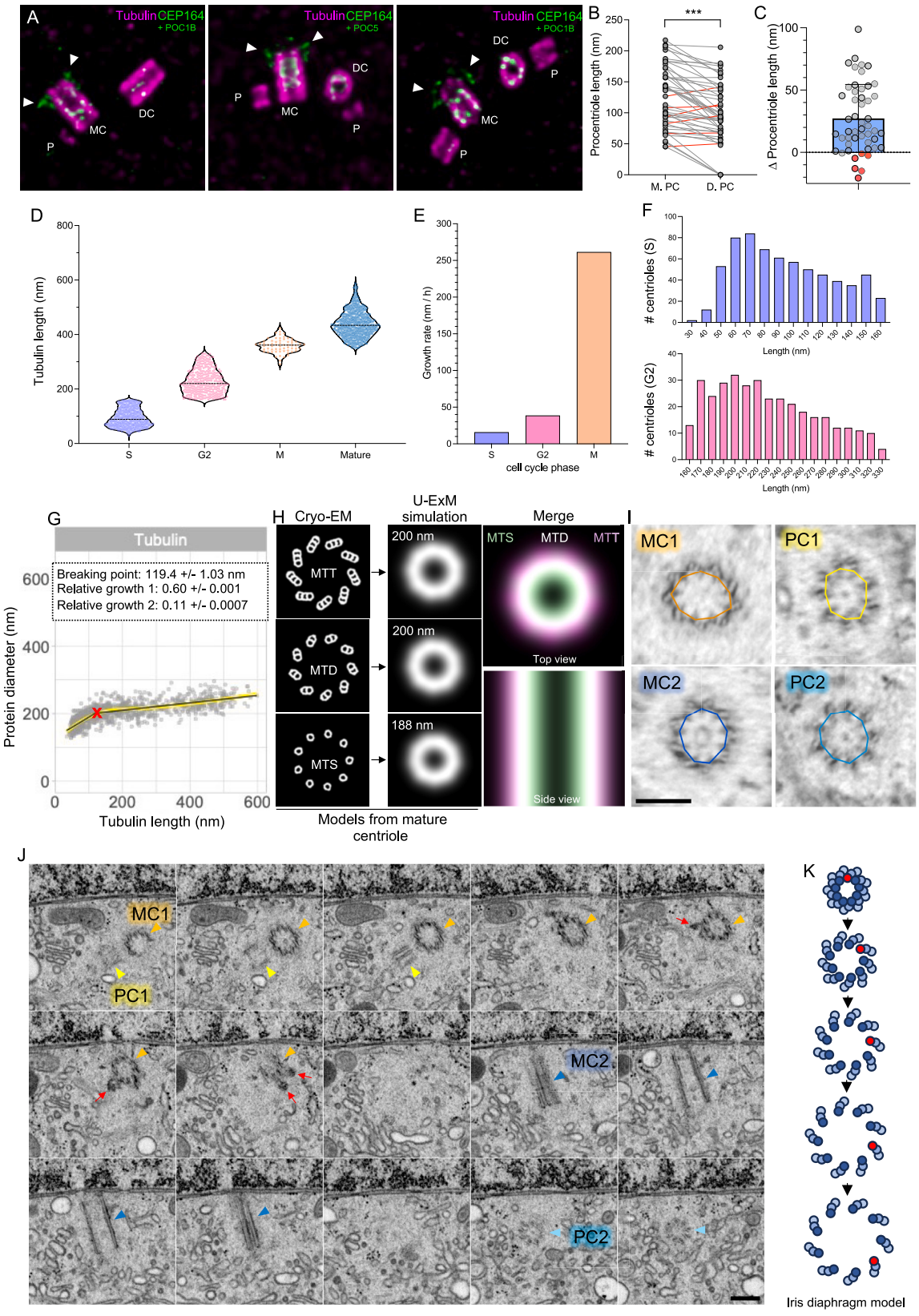
(L) Overlapping radial and longitudinal positions of the pinhead components, CPAP (mauve) and CEP44 (purple), relative to tubulin signal (gray). See Table S2 for statistics.

(M) Overlapping radial and longitudinal positions of the inner scaffold components centrin (dark gray), POC5 (yellow), FAM161A (green), and POC1B (blue) relative to tubulin signal (gray). See Table S2 for statistics.

(N and O) Centrioles from U2OS cells stained for tubulin and γ-tubulin, observed from side view (N) and from top view (O). γ-tubulin is located both in the lumen of the centriole and around the centriole, with two distinct localizations in the proximal (white arrowheads) and distal (yellow arrowheads) regions. Scale bars: 200 nm (N) and 100 nm (O).

(P) Overlapping radial and longitudinal positions of the distal cap components CP110 (red), CEP290 (light red), CEP97 (yellow), CEP162 (pink), and CPAP (orange) relative to tubulin signal (gray). See Table S2 for detailed statistics.

(Q) Overlapping radial and longitudinal positions of the luminal distal ring components C2CD3 (light blue), SFI1 (light green), centrin (dark green), and CEP135 (dark blue) relative to tubulin signal (gray). See Table S2 for statistics.



(legend on next page)

Figure S4. Monitoring the growth of the centriolar microtubule wall, related to Figure 2

(A) Centrosomes from duplicating U2OS stained for tubulin and the distal appendages marker CEP164 (white arrowheads) and either POC1B or POC5 (inside the centrioles). As CEP164 is only present when the centriole reaches its final mature stage, it allows us to distinguish mother and daughter centrioles. Images show a systematic asymmetry of the procentriole length toward the mother centriole. Scale bars: 200 nm.

(B) Procentriole length growing from the mother (M. PC) or the daughter (D. PC) centriole showing a significant length asymmetry between the two procentrioles. ***$p < 0001$, paired Student's t test. $n = 50$ centrioles from >3 independent experiments.

(C) Procentriole length difference (delta) from each pair of centrosomes measured. Average ± SD: 27.2 ± 27.4 nm. $n = 50$ centrioles from >3 independent experiments.

(D–F) Procentriole growth estimation through cell-cycle phases. (D) Centriole length distribution through the different stages of the cell cycle (procentrioles during S phase, G2, mitosis, and mature centrioles), showing the constant increase in length. Given that POC5-positive procentrioles are typically found in the G2 phase,[74] procentrioles smaller than 160 nm, devoid of POC5 staining (Figure 5N), are thus presumed to be in the S phase. Average ± SD are as follow: S = 94.66 ± 32.73 nm, G2 = 228.4 ± 44.7 nm, M = 359.3 ± 25.64 nm, mature = 438.9 ± 47.42 nm. $n = 656$, 352, 62, and 798 centrioles for S, G2, M, and mature, respectively, from >5 independent experiments. (E) Growth rate of procentrioles during S phase, G2, and mitosis, based on data from (A). Growth rate in nm/h: S = 15.81 nm/h, G2 = 38.61 nm/h, and M = 261.4 nm/h. The difference between the maximum and minimum centriole length in each cell-cycle phase divided by the phase duration,[92] allows us to calculate the growth rate (nm/h), demonstrating a difference of growth rate between S, G2, and mitosis. (F) Histogram distribution of procentriole according to their length during S phase (purple) and G2 phase (pink). $n = 656$ (S) and 352 (G2) centrioles per condition.

(G) Plot of loess fit (yellow) compared with a segmented linear regression (black line), with one breakpoint showing a phase of diameter and length growth at a near-linear rate (slope of 0.60 ± 2 × 0.0005). This phase, called bloom phase, stops when tubulin length reaches 119 nm and a second phase of elongation starts at a lower rate (slope of 0.11 ± 2 × 0.0003). In the box are the breakpoint location (red cross) and the slopes (relative growth) of the two segments, along with their respective standard errors. The superposition shows that the segmented regression is a good representation of the overall growth.

(H) Model of a *Chlamydomonas* mature centriole extracted from a cryo-EM image, represented with triplet (top), doublet (middle), or singlet (bottom) microtubules. Images were scaled down to mimic a fluorescent signal of expanded centrioles, and the diameters of each representation were measured. Right panel shows the merge of the three representations in top view and side view (MTT, microtubule triplet, pink; MTD, microtubules doublet, white; MTS, microtubule singlet, green).

(I) Extracted images from the raw image (see D) of mature centrioles (MC1 and MC2) and procentrioles (PC1 and PC2) in top views. Each image is superimposed with the nine-sided polygon used to calculate the mean diameter of the ellipse passing through each A-microtubule (see Figure 2G). Scale bars: 200 nm.

(J) Image montage of the raw FIB-SEM (from https://openorganelle.janelia.org/datasets/jrc_hela-2) in the centrosome region, allowing us to visualize the two mature centrioles (MC1, orange; MC2, blue) with their respective procentrioles (PC1, yellow; PC2, blue). Note the presence of distal appendages (red arrows) on the mature centriole 1, indicating that it is the mother. Scale bar: 200 nm.

(K) Scheme of the iris diaphragm model from Albrecht-Buehler.[57]

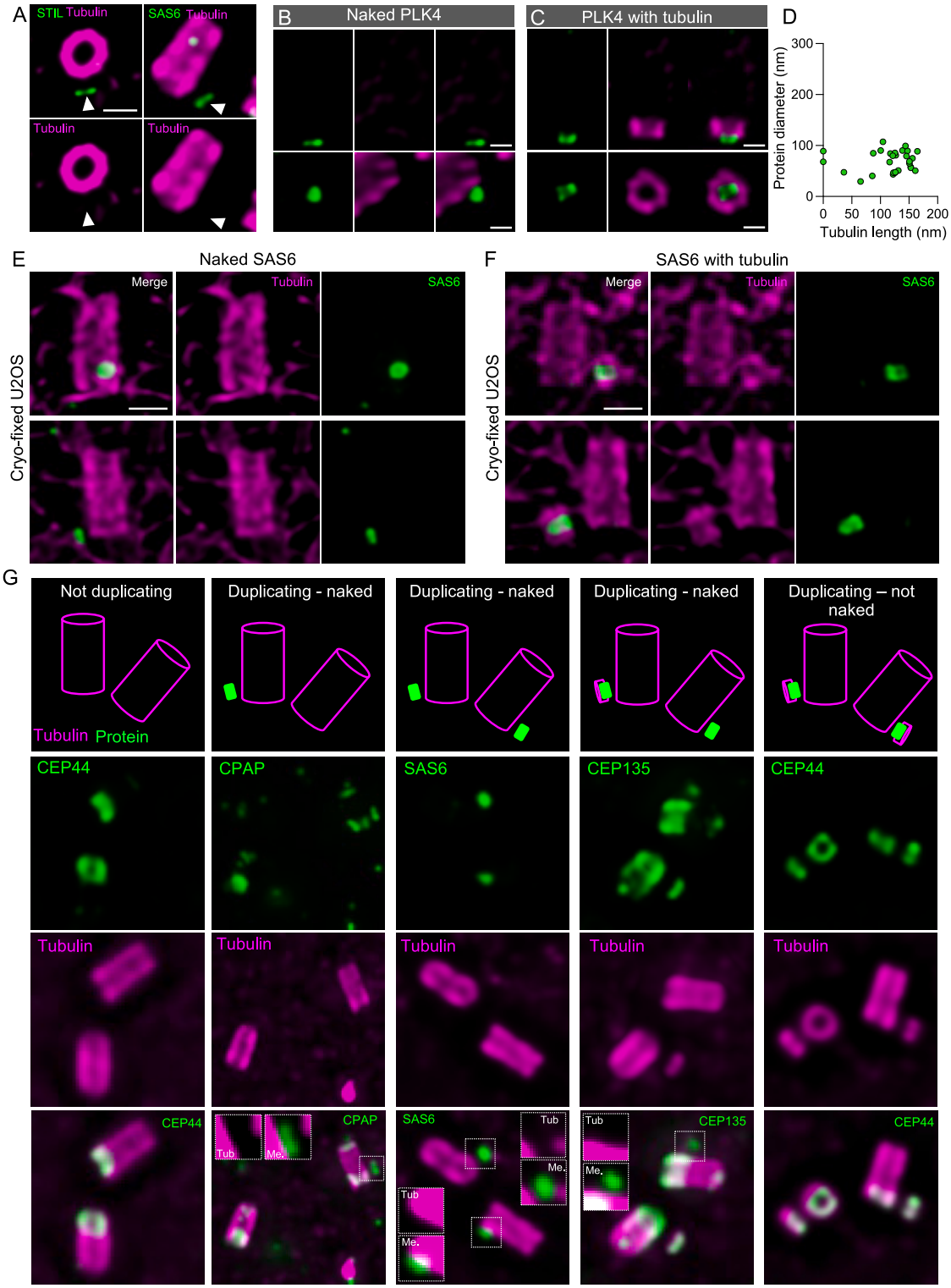


Figure S5. Methodology of the naked cartwheel observation, related to Figure 3

(A) Overexposed image presented in Figures 3A and 3B showing the presence of STIL (left, white arrowhead) and SAS-6 (right, white arrowhead) in absence of tubulin signal. Scale bars: 200 nm.

(B and C) Procentrioles stained for tubulin (magenta) and PLK4 (green). Note that (B) shows the presence of PLK4 at the level of the future centriole without the tubulin signal (naked), and (C) shows the early procentriole where the PLK4 signal can be found at the level of a tubulin-positive procentriole. Scale bars: 100 nm.

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(D) PLK4 diameter evolution during centriole assembly (relative to tubulin length), showing no increase in PLK4 diameter.
(E and F) Centrioles from cryo-fixed U2OS cells stained for tubulin and SAS-6. (E) shows the presence of SAS-6 at the level of the future centriole without the tubulin signal, and (C) shows the early procentriole where SAS-6 signal is found at the level of a tubulin-positive procentriole. Scale bars: 200 nm.
(G) Examples of widefield images used for the quantification of the percentage of cell with naked proteins (Figure 3C). Top panel shows a graphical representation of each situation ("not duplicating," "duplicating—naked," and "duplicating not naked"). Middle and bottom panels show representative images of each situation. Dashed squares indicate overexposed zoom in, showing the absence of tubulin in the naked situation. Scale bars: 250 nm.

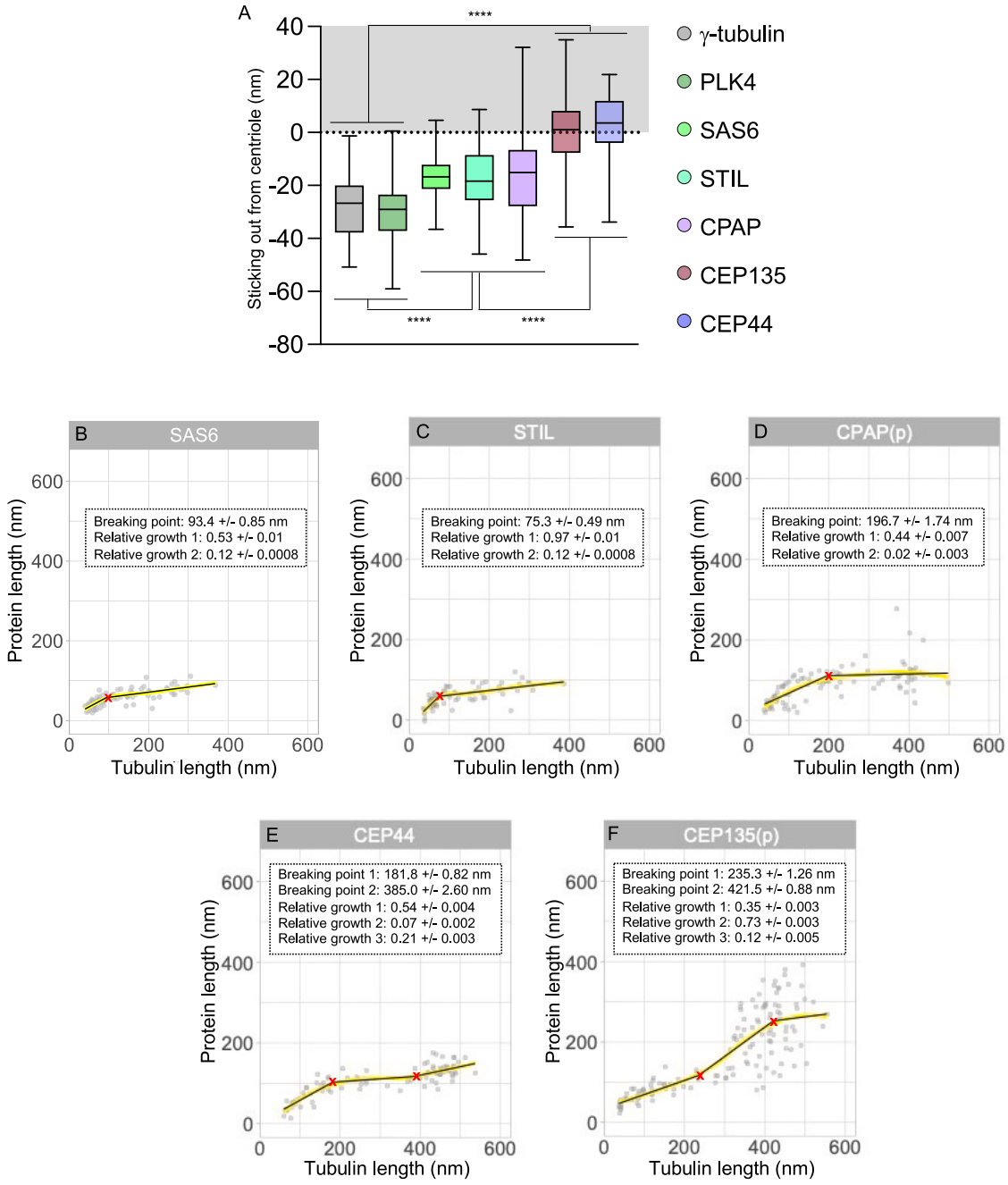


Figure S6. Growth rate of the cartwheel and pinhead components, related to Figure 4

(A) Distance between protein signal and tubulin signal start (sticking out). Average ± SD are as follows: PLK4 = −30.59 ± 12.5 nm, γ-tub = −28.25 ± 11.2 nm, SAS-6 = −16.72 ± 8.2 nm, STIL = −17.72 ± 11.3 nm, CPAP = −13.93 ± 16.6, CEP135 = −0.37 ± 13.7 nm, and CEP44 = 2.36 ± 11.6 nm. n = 38, 28, 79, 81, 58, 57, and 41 centrioles for γ-tub, PLK4, SAS-6, STIL, CPAP, CEP135, and CEP44, respectively, from three independent experiments. Statistical differences were assessed using a one-way ANOVA followed by Tukey's post hoc test. ****$p < 0.0001$ in all conditions tested.

(B–F) Plots of loess fits and superimposed segmented regressions per protein showing that the segmented regressions are a good representation of the overall data (as in Figure S4D).

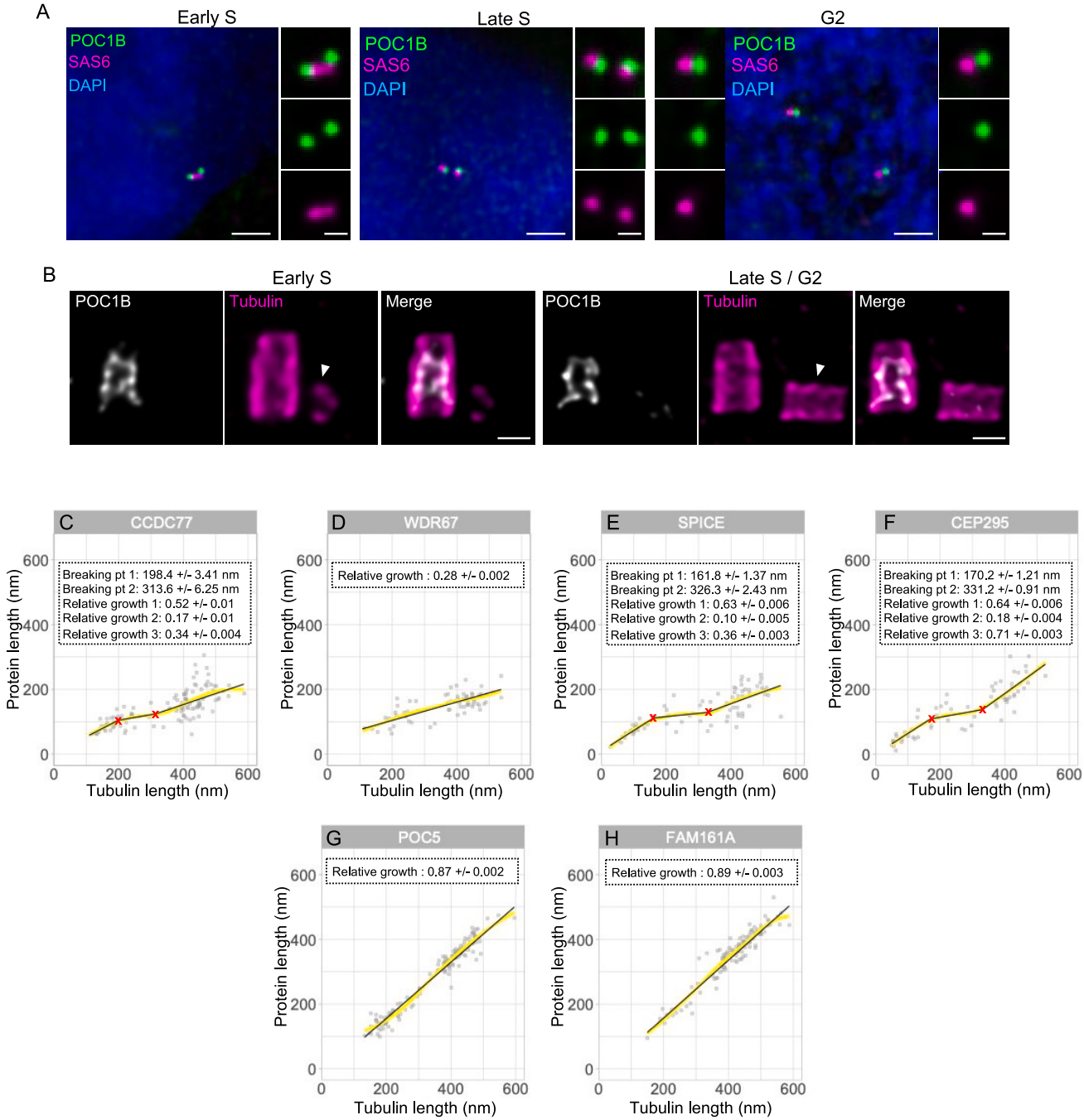


Figure S7. Characteristics of the MTT connectors, related to Figure 5

(A) Non-expanded U2OS stained for POC1B (green; dilution 1:500), SAS-6 (magenta; dilution 1:100), and DAPI (blue) during cell-cycle progression. Images show no colocalization between SAS-6 (procentriole marker) and POC1B, indicating that POC1B is only present at mature centrioles. Scale bars: 2 μm and 500 nm (inset).

(B) Centrioles from expanded U2OS stained for tubulin and POC1B during cell-cycle progression. Note that expansion confirmed that POC1B is only present at the level of mature centrioles and not at procentrioles. Scale bars: 200 nm.

(C–F) Plots of loess fits and superimposed segmented regressions per protein showing that the segmented regressions are a good representation of the overall data (as in Figure S4D).

(G and H) Plots of loess fits and superimposed segmented regressions per protein, showing that the segmented regressions are a good representation of the overall data (as in Figure S4D).

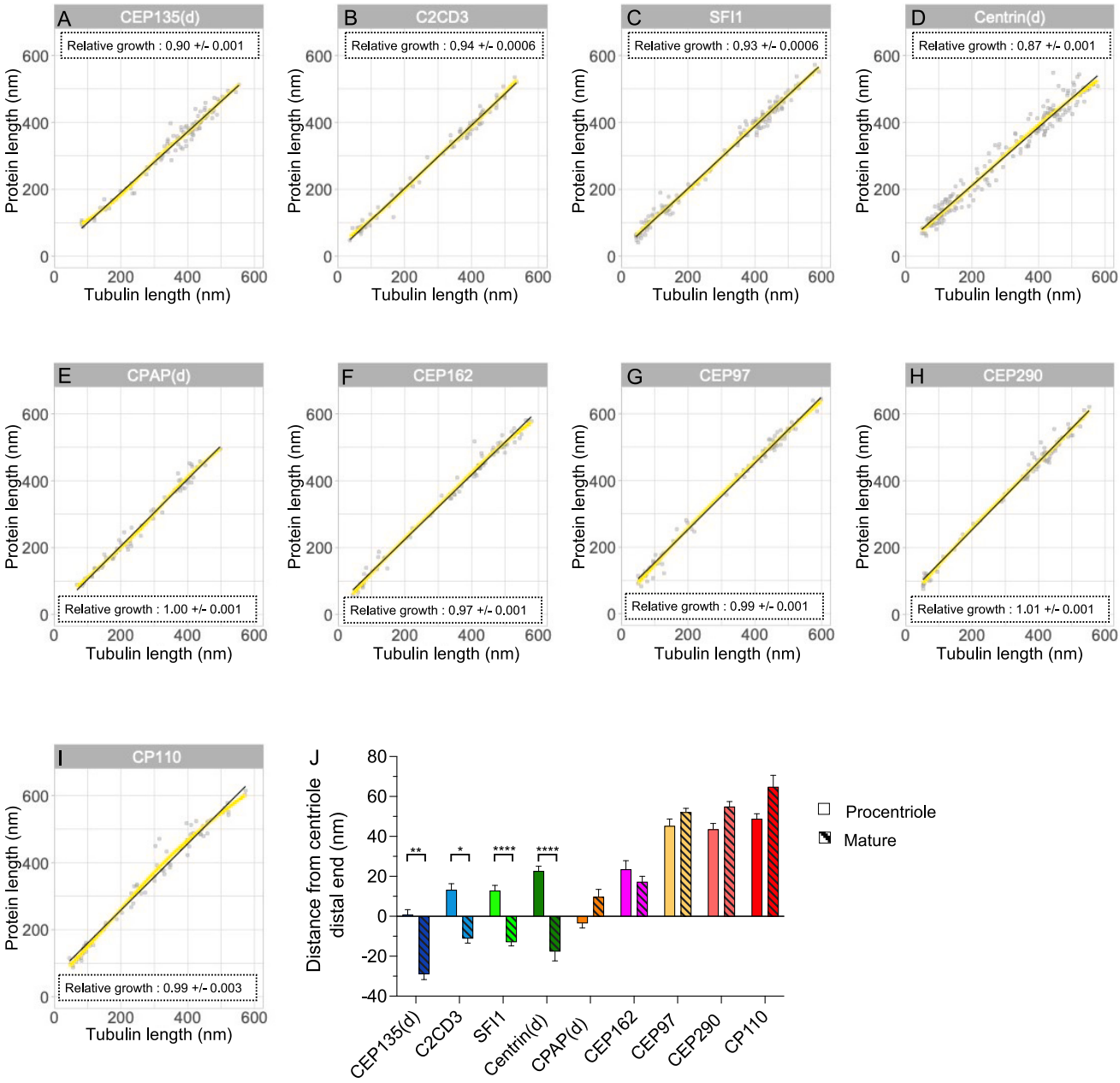
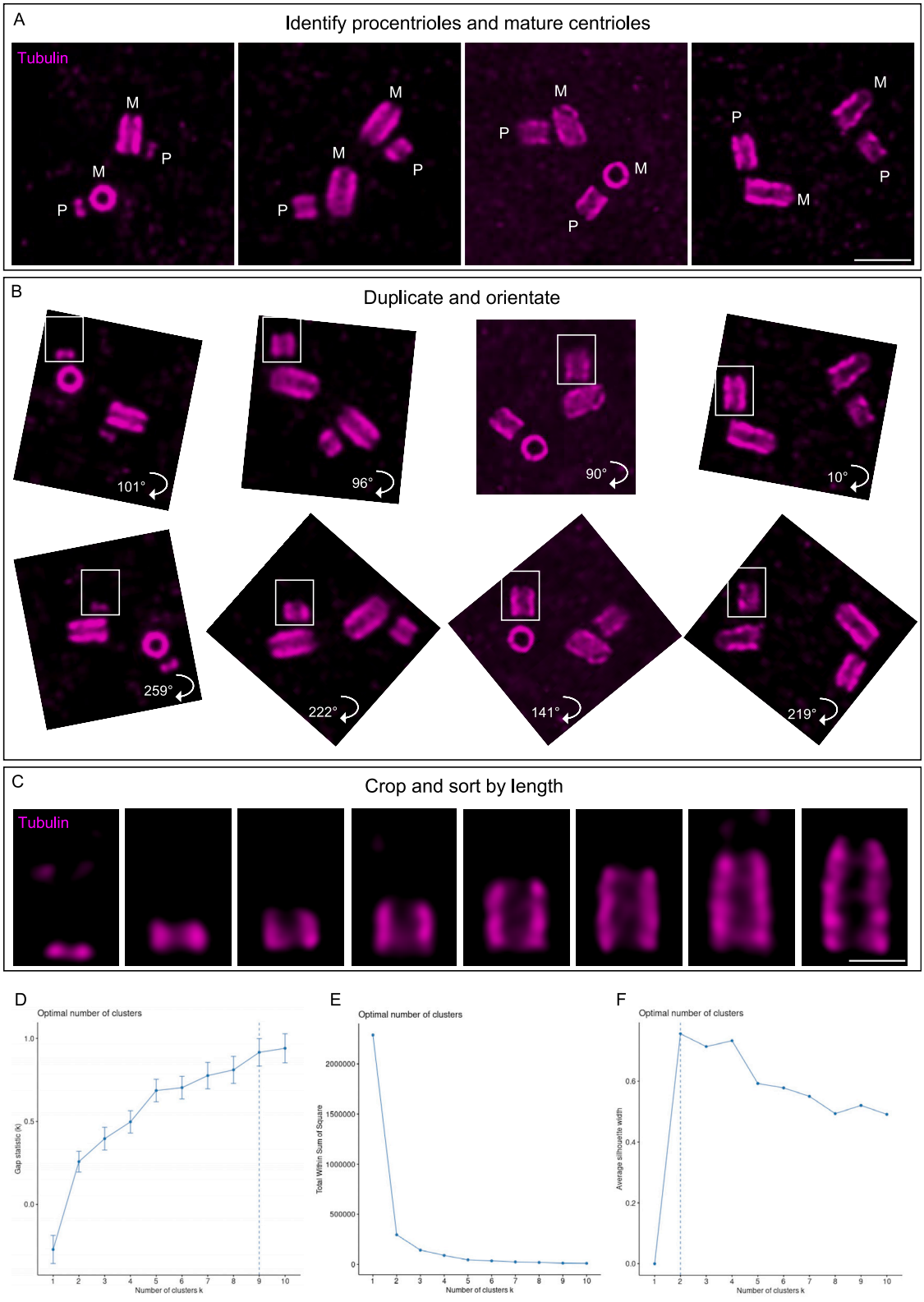


Figure S8. Characteristics of the distal proteins, related to Figure 6

(A–I) Plots of loess fits and superimposed segmented regressions per protein, showing that the segmented regressions are a good representation of the overall data (as in Figure S4D).

(J) Distance between the indicated proteins and the end of the centriole, assessed by the tubulin signal. Negative values indicate that the protein ends before the end of the centriole, while positive values indicate that the protein sticks out from the centriole. Plain bars depict the measurements done on procentrioles, striped bars depict measurements on mature centrioles. CEP135(d) procentriole vs. mature centriole: $^{**}p = 0.002$; C2CD3 procentriole vs. mature centriole: $^{*}p = 0.012$; SFI1 procentriole vs. mature centriole: $^{***}p < 0.0001$; centrin(d) procentriole vs. mature centriole: $^{***}p < 0.0001$, Kruskal-Wallis followed by Dunn's post hoc test. See Table S2 for statistics.



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Figure S9. Data representation, related to STAR Methods

(A) Z-projection of images showing duplicating centrosomes at different stages of procentriole assembly (M, mature; P, procentriole). Note that images with non-tilted centrioles and procentrioles were selected. Scale bars: 500 nm.

(B) Duplication and rotation of the images shown in (A) to orientate the centrioles from proximal to distal.

(C) 50 × 75 pixel crops of the images shown in (B), sorted by centriolar length to create the assembly frieze. Scale bars: 200 nm.

(D–F) Commonly used criteria for the selection of the number of clusters. (D) The GAP statistic that maximizes the non-uniform distribution of observations. (E) The elbow statistic (also known as WSS or within sums of squares) that seeks to identify when adding another cluster does not improve the overall clustering quality (as measured by WSS). (F) The silhouette method that looks for maximizing cluster homogeneity. Joint consideration of these plots, of Figure 7, and the domains (range of values) of growth of the set of proteins lead to the choice of k = 7 clusters. More clusters do not bring any advantage in terms of overall cluster quality, but less would not take into account biological characteristics such as the maximum and type of growth (linear for inner scaffold, distal cap, luminal distal ring, or nonlinear for cartwheel, pinhead, and A-C linker).